# MATH5703M Statistical Theory

Jochen Voss

October 16, 2026

# Contents

**Introduction** **vii**

**1 Statistical Models and Estimation** **1**
1.1 Statistical Models . . . 1
1.2 Statistics and Estimators . . . 3

**2 Bias, Mean Squared Error and Consistency** **6**
2.1 Bias and Unbiasedness . . . 6
2.2 Mean Squared Error . . . 8
2.3 When Unbiasedness Is Not Enough . . . 9
2.4 Consistency . . . 10

**3 R Basics and a First Simulation** **13**
3.1 First Steps in R . . . 13
3.2 Simulating a Normal Sample . . . 14
3.3 The Sample Mean as a Random Quantity . . . 15

**4 Likelihood and the Method of Moments** **18**
4.1 The Likelihood Function . . . 18
4.2 The Score Function . . . 20
4.3 The Method of Moments . . . 20
4.4 Identifiability . . . 22

**5 Maximum Likelihood Estimation** **25**
5.1 The Maximum Likelihood Estimator . . . 25
5.2 Worked Examples . . . 26
5.3 When the Recipe Fails . . . 28
5.4 Invariance Under Reparametrisation . . . 29

**6 Estimators by Simulation** **33**
6.1 Functions and Loops . . . 33
6.2 Empirical Bias and Mean Squared Error . . . 34
6.3 Comparing Two Estimators . . . 36

**7 Exponential Families** **39**
7.1 Recognising the Shape . . . 39
7.2 Several Parameters . . . 41
7.3 The Natural Parametrisation and the Cumulant Function . . . 42
7.4 What the Structure Buys . . . 43

**8 Sufficiency** **48**
8.1 Sufficient Statistics . . . 48
8.2 The Factorisation Theorem . . . 49
8.3 Sufficiency in Exponential Families . . . 51
8.4 The Sample Maximum and Minimum . . . 51
8.5 Minimal Sufficiency and Completeness . . . 52

**9 Consistency by Simulation** **56**
9.1 A Running Estimate . . . 56
9.2 The Sampling Distribution as $n$ Grows . . . 57
9.3 The Mean Squared Error as a Function of the Sample Size . . . 59
9.4 A Sufficient Statistic in R . . . 61
9.5 A Task to Take Home . . . 62

**10 Improving Estimators: Rao–Blackwell and Lehmann–Scheffe** **64**
10.1 The Rao–Blackwell Theorem . . . 64
10.2 Completeness and the Lehmann–Scheffe Theorem . . . 65
10.3 Ancillary Statistics and Basu's Theorem . . . 67

**11 Fisher Information** **70**
11.1 Setting and Assumptions . . . 70
11.2 The Score Has Mean Zero . . . 71
11.3 Fisher Information . . . 71
11.4 Examples . . . 73
11.5 Exponential Families . . . 73
11.6 Reparametrisation . . . 74
11.7 When the Identities Fail . . . 75

**12 Sampling Distributions and Standard Errors** **78**
12.1 The Sampling Distribution of an Estimator . . . 78
12.2 Standard Errors and a Wald Interval . . . 79
12.3 Estimating a Probability by Simulation . . . 80
12.4 Skewed Data . . . 81

**13 The Cramer–Rao Inequality** **84**
13.1 The Inequality . . . 84
13.2 Estimating a Function of the Parameter . . . 86
13.3 Efficiency and Attainment . . . 86
13.4 When the Conditions Fail . . . 88

**14 Multiparameter Fisher Information** **91**
14.1 The Score Vector and the Information Matrix . . . 91
14.2 Properties of the Information Matrix . . . 92
14.3 The Cramer–Rao Bound for Vector Parameters . . . 93
14.4 The Normal Distribution . . . 94
14.5 Nuisance Parameters . . . 95

**15 Cramer–Rao and Efficiency in R** **98**
15.1 An Efficient Estimator Against the Bound . . . 98
15.2 Attaining and Missing the Bound . . . 99
15.3 Beating the Bound Without Breaking It . . . 100
15.4 A Task to Take Home . . . 102

**16 Consistency of the MLE** **104**
16.1 The Regularity Conditions . . . 104
16.2 The Likelihood Ratio Detects the Truth . . . 105
16.3 Consistency . . . 106
16.4 When the Assumptions Fail . . . 107

**17 Asymptotic Normality and Efficiency of the MLE** **111**
17.1 The Theorem . . . 111
17.2 Proof of the Theorem . . . 112
17.3 Standard Errors . . . 113
17.4 Examples . . . 114
17.5 Quality of the Approximation . . . 115

**18 Maximum Likelihood in R** **118**
18.1 The Log-Likelihood as an R Function . . . 118
18.2 Standard Errors from the Observed Information . . . 119
18.3 Starting Values . . . 120
18.4 Coverage of the Wald Interval . . . 121
18.5 A Task to Take Home . . . 123

**19 The EM Algorithm** **125**
19.1 Incomplete Data . . . 125
19.2 The Algorithm . . . 126
19.3 The Ascent Property . . . 127
19.4 A Two-Component Normal Mixture . . . 128
19.5 Practical Matters . . . 130

**20 Hypothesis Testing** **132**
20.1 Hypotheses . . . 132
20.2 Tests, Critical Regions and Errors . . . 132
20.3 The Power Function, Size and Level . . . 133
20.4 Testing the Mean of a Normal Sample . . . 134
20.5 The $p$-Value . . . 135

**21 EM in R** **139**
21.1 The E and M Steps as R Functions . . . 139
21.2 A Larger Sample . . . 140
21.3 Label Switching and a Bad Start . . . 143
21.4 A Task to Take Home . . . 144

**22 The Neyman–Pearson Lemma** **146**
22.1 Simple Hypotheses and Most Powerful Tests . . . 146
22.2 The Lemma . . . 147
22.3 The Normal Mean . . . 148

22.4 One-Sided Alternatives . . . 149
22.5 Two-Sided Alternatives . . . 150

**23 Likelihood-Ratio Tests** **153**
23.1 The Generalised Likelihood Ratio . . . 153
23.2 The Two-Sided Test for the Mean of a Normal Distribution . . . 154
23.3 Wilks' Theorem . . . 155
23.4 Using the Theorem . . . 156

**24 Power and Size by Simulation** **160**
24.1 The Size of a Test by Simulation . . . 160
24.2 The Power Curve . . . 161
24.3 Wilks' Theorem in Action . . . 162
24.4 A Task to Take Home . . . 165

**25 Confidence Intervals** **167**
25.1 Definition and Interpretation . . . 167
25.2 Pivotal Quantities . . . 168
25.3 Inverting a Test . . . 169
25.4 Approximate Intervals from the MLE . . . 170
25.5 One-Sided Intervals and the Choice of Interval . . . 171

**26 Bayesian Inference: Priors and Posteriors** **174**
26.1 Prior, Likelihood and Posterior . . . 174
26.2 Conjugate Priors . . . 175
26.3 Point Estimates from the Posterior . . . 176
26.4 Flat, Improper and Jeffreys Priors . . . 177

**27 Confidence Versus Credibility** **181**
27.1 Coverage by Simulation . . . 181
27.2 The Same Data, Two Intervals . . . 183
27.3 From Prior to Posterior . . . 184

**28 Credible Intervals and Bayesian Testing** **187**
28.1 Credible Intervals . . . 187
28.2 Bayesian Testing . . . 188
28.3 Comparing Frequentist and Bayesian Answers . . . 190

**29 Standard Tests Revisited** **194**
29.1 Tests from Pivots . . . 194
29.2 The One-Sample $t$-Test . . . 195
29.3 The Chi-Squared Test for a Variance . . . 196
29.4 Paired and Two-Sample Problems . . . 197
29.5 Which Test? . . . 198

**30 Do the Standard Tests Work?** **201**
30.1 Real Data, Two Answers . . . 201
30.2 The Replication Crisis in Miniature . . . 202
30.3 A Stress Test . . . 205
30.4 A Task to Take Home . . . 206

**31 The Exponential Model: Start to Finish** **208**
31.1 The Model and the Data . . . 208
31.2 Point Estimation . . . 209
31.3 Interval Estimation . . . 209
31.4 Testing . . . 210
31.5 The Bayesian Answer . . . 210
31.6 Looking Back and Looking Forward . . . 211

**Preparing for the Exam** **215**

**A Background from Probability** **223**
A.1 Expectation, Variance and Independence . . . 223
A.2 Standard Distributions . . . 224
A.3 Sampling from a Normal Population . . . 225
A.4 Some Inequalities . . . 226
A.5 Conditional Expectation . . . 227
A.6 Convergence and Limit Theorems . . . 228
A.7 Transformation of a Random Variable . . . 230
A.8 Bayes' Theorem . . . 230

**B Introduction to R** **231**
B.1 Getting Started . . . 231
B.2 Vectors and Arithmetic . . . 232
B.3 Lists . . . 234
B.4 Random Numbers and Distributions . . . 235
B.5 Writing Your Own Code . . . 236
B.6 Estimation by Simulation . . . 238
B.7 Plotting, and Getting Results into a Report . . . 240
B.8 Numerical Maximum Likelihood . . . 241

**C Solutions to the Exercises** **243**
C.1 Statistical Models and the Estimation Problem . . . 243
C.2 Bias, Mean Squared Error and Consistency . . . 244
C.3 Likelihood and the Method of Moments . . . 246
C.4 Maximum Likelihood Estimation . . . 248
C.5 Exponential Families . . . 252
C.6 Sufficiency . . . 257
C.7 Rao–Blackwell and Lehmann–Scheffe . . . 260
C.8 Fisher Information . . . 264
C.9 The Cramer–Rao Inequality . . . 268
C.10 Multiparameter Fisher Information . . . 272
C.11 Consistency of the MLE . . . 274
C.12 Asymptotic Normality of the MLE . . . 276
C.13 The EM Algorithm . . . 277
C.14 Hypothesis Testing . . . 280
C.15 The Neyman–Pearson Lemma . . . 282
C.16 Likelihood-Ratio Tests . . . 285
C.17 Confidence Intervals . . . 288
C.18 Bayesian Inference: Priors and Posteriors . . . 291
C.19 Credible Intervals and Bayesian Testing . . . 294

C.20 Standard Tests Revisited . . . . . 298
C.21 The Exponential Model: From Start to Finish . . . . . 301

**Bibliography** **306**

**Index** **308**

# Introduction

These are the lecture notes for the Statistical Theory (MATH5703M) module. The question we will consider in this module, in many different forms, is the following: What can we learn about the parameters of a model if we are given data from the model?

The first half of the module is concerned with estimation. We will start by considering statistical models and properties of an estimator, using the bias and mean squared error as measures of quality. We will then consider two different approaches to constructing estimators: maximum likelihood estimators and the method of moments. The unifying framework for this part of the module is the exponential family of distributions. Sufficiency, completeness and the Rao–Blackwell and Lehmann–Scheffe theorems will be used to study how to improve an estimator and to identify cases where an estimator cannot be improved. Finally, we will consider Fisher information and the Cramer–Rao bound to study limitations on the precision of unbiased estimators. We will conclude the study of estimation by considering large-sample theory for the maximum likelihood estimator and the EM algorithm.

The second half of the module is concerned with testing and interval estimation. We will develop the Neyman–Pearson theory of hypothesis tests and the likelihood-ratio test, and will then consider confidence intervals. A section on Bayesian inference will cover the choice of prior, credible intervals and Bayesian testing. We will conclude the study of testing by considering standard tests used in practice.

Alongside the theory, we will use simulation in R to illustrate the ideas of the module. Appendix B collects the R commands we use.

## How to Read the Proofs

Many of the benefits of this module can be gained by studying the proofs, but a different approach is needed to read and understand a proof than is needed to understand ordinary text. The technique we will use here, called *self-explanation*, is the only technique for reading proofs which is supported by strong experimental evidence. In three experiments with undergraduate mathematics students, Hodds et al. (2014) showed that a short self-study training in self-explanation, consisting of around twenty minutes of slides and one worked example and one practice proof, significantly improved scores in proof-comprehension tests. The improvement in the laboratory experiment was close to one standard deviation. A fifteen-minute booklet version of the training, completed during a scheduled lecture, also showed a smaller but significant improvement, and the effect of this was still visible three weeks later. We will practise this technique together in

the first week of the module, and my hope is that you will use the technique throughout the module.

The idea of self-explanation is that you should pause after every line of the proof, and try to explain the line to yourself, before you read the next line. A good self-explanation answers the following three questions.

- What does this line state?
- Why is this statement true? Be specific about how you know: is it from an earlier line of the proof, or from a definition, or from an assumption of the theorem, or from a previous result?
- How does this line contribute to the proof? How does it move the argument towards the statement?

There are two bad habits which look similar to self-explanation, but are not: paraphrasing, *i.e.* restating a line in different words without understanding the reason, and 'monitoring', *i.e.* saying 'yes, this makes sense' and then carrying on. Both of these techniques help your understanding less than self-explanation does. It is not enough to just have a feeling that you understand a proof; you also need to be able to reconstruct the argument line by line. If you cannot name the reason for a statement, you have a gap in your understanding and you need to find it: write down exactly what is missing, look it up or work it out, and only then continue.

In practice this means reading with a pencil. Before you read the proof, read the statement carefully, and make sure you understand what is assumed and what is claimed. Then, as you read the proof, write down the justification for each step in the margin. If you cannot find a justification for a step, try to fill the gap yourself before you read the next line. This approach will make reading a proof slower, but it will also make sure that you understand the proof. The same technique can be used to understand the worked solutions in appendix C: try to solve the exercise yourself first, and only then look at the solution line by line to check your work.

## What About AI?

Large language models are very good at the kind of material we cover in this module. Using one of these models, it is possible to get a solution to any of the exercises in these notes, to get answers for the examination questions, and even to get R code for most of the tasks required for the practical report.

The problem with this is that the aim of the module is for you to learn the material in these notes, not to solve the exercises (solutions for all of the exercises are in the appendix). While AI can explain every line of a proof to you, this is not the same as self-explanation. Similarly, the practical report requires you to show your judgement and to explain your work, but a report written by the model does not give you practice in either of these skills.

There is evidence of this problem in scientific studies: Strömberg et al. (2026) studied the effect of such tools on 26 811 secondary-school students over a period of 30 months: homework scores increased by 18%, homework time decreased by 30% and after six months scores in the monthly, closed-book examinations had

decreased by 20%. The losses were concentrated amongst the approximately 80% of users who completed their homework faster than any non-user and scored high marks for the homework, and the authors of the study interpreted these results as indicating that these users had "outsourced" their homework to the tool. Users who spent as much time on their homework as non-users did performed about as well in the examinations, so the loss in this group was not due to the tools themselves, but due to the way the tools were used.

A randomised trial in mathematics (Bastani et al., 2025) showed the same effect. Around 1000 secondary-school students either used a language model to learn the material or used only the textbook and notes. The students were then tested in a closed-book exam. The students who used the unrestricted language model scored 17% worse than the others, mainly because they used the model to find answers to the questions. When a version of the model was used which only gave hints instead of answers, the loss in performance disappeared. The students were not aware of the difference: students who used the unrestricted model did not believe that they had learned less than the other students.

I want to also mention that the exam for our module is closed book, pen and paper based. In the exam, you will not have access to AI support, and will need to rely on what you have personally learned.

There are ways to use these models to help understanding, but these require a slightly different approach: you ask the model for something other than the answer to the question at hand.

- Ask for more practice material. The model can easily generate more exercises for you. But remember that the only way to learn from these exercises is to solve them yourself!

- Ask for a second explanation. If a definition, a result or a step in a proof is not clear to you, ask the model to explain it in different words; sometimes a second explanation helps you to understand it. Afterwards, close the window and write it down from memory: a definition or result in full, a step in a proof by giving the reason why it holds in your own words. Reading the model's explanation is only monitoring, while writing it down from memory shows that you have understood or at least memorised it.

- Ask which result a step relies on. If you are stuck on a step of a proof which you cannot yet justify, the model can sometimes name the result the step uses, for example Jensen's inequality or the tower property of conditional expectation. Ask for the name only, look up the result in the index and work out the step yourself.

- Ask why, not what. A question like "Why do we divide by $n - 1$ in the sample variance?" is a question which you can first try to answer yourself, and then check your answer with the model's answer. In contrast, a question like "solve this exercise" leaves nothing for you to do.

- Ask about R. What does an error message mean? What does a given argument of a command do? These are questions which can be answered by the help pages (see appendix B). It is not expected that you will know these details by heart from this module, so asking the model is fine.

One way to test whether you have understood an argument is to try to re-create the argument on paper, from memory, a week later. If you can do this, you have learned something! If you cannot do this, you only have a solution, not an understanding.

## Reading

These notes are self-contained, but the following books can be used as additional reading. The links lead to the University of Leeds Library catalogue.

- G. Casella and R. L. Berger, *Statistical Inference* (2nd edition; available online). This is our main text. It covers most of the material of the module, from point estimation to sufficiency and the information inequality, and from hypothesis testing to confidence intervals.
- G. A. Young and R. L. Smith, *Essentials of Statistical Inference* (print only). A concise and rigorous graduate text at the level of this module. The book is particularly good on likelihood, decision theory and the large-sample behaviour of the maximum likelihood estimator.
- G. H. Givens and J. A. Hoeting, *Computational Statistics* (2nd edition; available online). This book is useful for learning about the EM algorithm and about numerical optimisation of the likelihood. It also introduces the computational ideas which we use in the R sessions.
- P. M. Lee, *Bayesian Statistics: An Introduction* (4th edition; available online). A self-contained introduction to the Bayesian material we cover, including the choice of prior, Jeffreys priors and Bayesian testing.
- W. J. Braun and D. J. Murdoch, *A First Course in Statistical Programming with R* (3rd edition; available online). This book is designed to support the R sessions and the practical report. It teaches programming and simulation in R from the ground up.

The following list has additional resources which either are more specialised and go deeper, or offer a gentler alternative to the more standard texts listed above.

- D. R. Cox and D. V. Hinkley, *Theoretical Statistics* (available online). The classic deeper treatment of the theory behind the module.
- E. L. Lehmann and G. Casella, *Theory of Point Estimation* (2nd edition; print only). The advanced reference for the estimation half: sufficiency, completeness and efficiency in full detail.
- A. C. Davison, *Statistical Models* (print only). Broad and likelihood-centred, with the EM algorithm and many worked applications.
- A. Gelman et al., *Bayesian Data Analysis* (3rd edition; available online). The standard modern reference for Bayesian statistics, well beyond what we need but the natural next step.

- M. H. DeGroot and M. J. Schervish, *Probability and Statistics* (4th edition; available online). A gentle and thorough account, useful for revising the prerequisite material from MATH2701.
- J. A. Rice, *Mathematical Statistics and Data Analysis* (3rd edition; print only). An accessible companion pitched a little below the core, good for a second explanation.

# Lecture 5

# Maximum Likelihood Estimation

In the previous lecture we have seen that for the exponential model, the parameter value which maximises the likelihood is the one which 'best' explains the data. In this lecture we will make this idea into the main estimation method of the module: we define the maximum likelihood estimator, practise computing it for the families from tables 1.1 and 1.2, and will see problems where more thought is needed. We will see that the estimator is not unbiased in general, but is invariant under reparametrisation; the deeper justification of the method will be given in lectures 13, 16 and 17.

## 5.1 The Maximum Likelihood Estimator

Example 4.5 already contains the idea: for the exponential sample the log-likelihood increases up to $\theta = 1/\bar{x}$ and then decreases, so this value fits the data best. We now formalise this principle.

**Definition 5.1.** A *maximum likelihood estimator* (MLE) for $\theta$ is an estimator $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$, which maximises the likelihood function over the parameter space, *i.e.* which satisfies

$$L(\hat{\theta}) \geq L(\theta) \qquad \text{for all } \theta \in \Theta.$$

Since the logarithm is strictly increasing, $\hat{\theta}$ maximises $L$ if and only if it maximises the log-likelihood $\ell$ and for the reasons given in lecture 4 we mostly work with $\ell$. The definition is careful to avoid the problem of a maximum not existing or not being unique. For example, the likelihood could increase as the edge of the parameter space is approached, so that no maximum exists. Similarly, the maximum may not be unique. In such cases, any maximiser can be used. For the standard families of models and typical data, neither of these problems will occur and it will be safe to speak of "the" MLE.

If the set $\Theta$ is an interval and the log-likelihood function is differentiable, then calculus can be used to find the MLE: at a maximum inside the interval $\Theta$, the derivative of $\ell$ equals zero and thus the MLE satisfies the *likelihood equation*

$$\ell'(\theta) = 0,$$

*i.e.* the MLE is a zero of the score function from definition 4.4. The general procedure is to find the log-likelihood function, to solve the likelihood equation and to verify that the solution is a maximum, for example by checking that $\ell''(\theta) < 0$ or by checking that the score changes from positive to negative.

*Warning.* The solution of the likelihood equation only gives *interior* critical points. These could be minima or inflection points and the checking step is required to avoid wrong answers. Also, the above procedure assumes that the maximum is interior and that $\ell$ is differentiable. We will see that these assumptions can be violated.

## 5.2 Worked Examples

We start our analysis by considering the MLE for commonly used families of probability distributions. For each of them we find the MLE using the recipe from the previous section, and then we check whether the resulting estimator is unbiased.

### The Exponential Distribution

The exponential distribution is the simplest of the three cases considered here: the likelihood equation has a single root and the moments of the MLE can be computed exactly.

**Example 5.2.** From examples 4.3 and 4.5 we know that the score function of an exponential sample $x_1, \dots, x_n$ is given by

$$\ell'(\theta) = \frac{n}{\theta} - \sum_{i=1}^{n} x_i$$

for all $\theta > 0$. Setting $\ell'(\theta) = 0$ gives $n/\theta = \sum_i x_i$, and thus the likelihood equation has the unique solution

$$\theta = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\bar{x}}.$$

To check that this root is a maximum, we differentiate once more and find

$$\ell''(\theta) = -\frac{n}{\theta^2} < 0$$

for all $\theta > 0$. Thus, the function $\ell$ is strictly concave and the root $1/\bar{x}$ is the only global maximum. We conclude that the MLE for the rate of the exponential distribution is

$$\hat{\theta} = \frac{1}{\bar{X}}.$$

This coincides with the method of moments estimator from example 4.7.

To find the bias of this estimator, let $T = X_1 + \cdots + X_n$, so that $\hat{\theta} = n/T$. In exercise 2.3 we found, using the gamma density of $T$, that $\mathbb{E}_\theta(\hat{\theta}) = n\theta/(n-1)$ for $n \geq 2$, while for $n = 1$ the expectation is infinite. Thus, for $n \geq 2$, the MLE is biased with bias

$$\operatorname{bias}(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta = \frac{n\theta}{n-1} - \theta = \frac{\theta}{n-1}.$$

Thus, the MLE for the rate is biased but asymptotically unbiased in the sense of definition 2.4. Multiplying $\hat{\theta}$ by $(n-1)/n$ removes the bias: the estimator $\tilde{\theta} = (n-1)\hat{\theta}/n = (n-1)/T$ satisfies $\mathbb{E}_\theta(\tilde{\theta}) = \theta$.

### The Bernoulli Distribution

For the Bernoulli distribution the recipe works whenever the data contain both successes and failures. The remaining two cases need separate treatment.

**Example 5.3.** Let $X_1, \ldots, X_n$ be an i.i.d. Bernoulli sample with success probability $\theta$, and let $s = \sum_i x_i$ be the observed number of successes. As in example 4.2 we have the likelihood $L(\theta) = \theta^s(1-\theta)^{n-s}$ and thus

$$\ell(\theta) = s \log \theta + (n-s)\log(1-\theta)$$

for all $\theta \in (0,1)$. Assume $0 < s < n$. Then the likelihood equation is

$$0 = \ell'(\theta) = \frac{s}{\theta} - \frac{n-s}{1-\theta}.$$

This equation implies $s(1-\theta) = (n-s)\theta$ and the only solution to this equation is $\theta = s/n$. Since $\ell''(\theta) = -s/\theta^2 - (n-s)/(1-\theta)^2 < 0$ everywhere, this root is the global maximum and the MLE for the proportion of successes is

$$\hat{\theta} = \frac{s}{n} = \bar{X}.$$

Since $\mathbb{E}_\theta(\bar{X}) = \theta$, the MLE for the Bernoulli parameter is unbiased.

*Remark.* The two cases $s = 0$ and $s = n$ are special. For $s = 0$ the function $\ell(\theta) = n\log(1-\theta)$ is strictly decreasing, and for $s = n$ the function $\ell(\theta) = n \log \theta$ is strictly increasing. In both cases the supremum is approached at the respective boundary point. We cannot find these points by setting $\ell'(\theta) = 0$. Additionally, in this module we use the open interval $\Theta = (0,1)$ as the parameter space, so for us no $\theta \in \Theta$ maximises the log-likelihood and if we are pedantic the MLE does not exist.

### The Normal Distribution

The normal distribution with both parameters unknown has a two-dimensional parameter vector, and thus the likelihood equation becomes a system of two equations.

**Example 5.4.** Consider $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where both parameters are unknown. Then we have $\theta = (\mu, \sigma^2)$. Taking logarithms we get

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

To find the maximum, we set both partial derivatives equal to zero. The partial derivatives are

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu)$$

and

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2.$$

The first derivative is zero only for $\mu = \bar{x}$. Substituting this value into the second derivative and setting it to zero gives $\sigma^2 = (1/n)\sum_i (x_i - \bar{x})^2$.

To check that this solution is the maximum, we note that for every fixed $\sigma^2$ the sum of squares $\sum_i (x_i - \mu)^2$ is smallest for $\mu = \bar{x}$, and thus $\ell(\mu, \sigma^2) \leq \ell(\bar{x}, \sigma^2)$. At $\mu = \bar{x}$ the derivative with respect to $\sigma^2$ can be written as $(\sum_i (x_i - \bar{x})^2 - n\sigma^2)/(2\sigma^4)$, which is positive to the left of the solution and negative to the right. Thus, the MLE is the pair

$$\hat{\mu} = \bar{X} \qquad \text{and} \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

The MLE $\hat{\mu} = \bar{X}$ for the mean is unbiased. The MLE $\hat{\sigma}^2$ for the variance is the estimator from example 2.5, where we found the bias $-\sigma^2/n$. Thus, $\hat{\sigma}^2$ is biased but asymptotically unbiased.

In all three examples considered in this section, the MLE is either unbiased or asymptotically unbiased. Note that there are other examples, where this property does not hold. The behaviour of the MLE for large samples in general is the subject of lectures 16 and 17.

## 5.3 When the Recipe Fails

The recipe for finding the MLE can fail in three different ways: the maximum may sit at a point where the log-likelihood is not differentiable, the likelihood equation may have several roots, and the likelihood equation may have no closed-form solution. We consider these three cases in turn.

### A Maximum at the Boundary

Both assumptions in the recipe, *i.e.* an interior maximum and differentiable log-likelihood, break down for the uniform model. In this case we have to return to definition 5.1 and maximise the expression directly.

**Example 5.5.** Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ be i.i.d., as in example 1.2. From the joint density found in exercise 1.3, the likelihood is

$$L(\theta) = \begin{cases} \theta^{-n} & \text{if } \theta \geq \max_i x_i, \\ 0 & \text{otherwise.} \end{cases}$$

Calculus is useless here: on the region where $L$ is positive its derivative $-n\theta^{-n-1}$ is negative everywhere, so the likelihood equation has no solution, and at $\theta = \max_i x_i$ the function is not even continuous. But maximising directly is easy: the likelihood is zero for $\theta < \max_i x_i$ and strictly decreasing for $\theta \geq \max_i x_i$, and thus largest at the left edge of the positive region. The MLE is the sample maximum,

$$\hat{\theta} = \max_i X_i.$$

This identifies the estimator we have been tracking since exercise 1.3, which by exercise 4.4 beats the method of moments estimator $2\bar{X}$ in mean squared error for every $n \geq 3$. The likelihood also rules out impossible parameter values automatically: for the dataset $x_1 = 0.1$, $x_2 = 0.2$, $x_3 = 0.9$ of lecture 4, any $\theta < 0.9$ has likelihood zero, so the contradiction the method of moments overlooked cannot arise.

### Several Roots

The uniform model fails the recipe in an obvious way. A failure which is harder to spot is a likelihood equation with several roots. Consider the $t$ location model with $\nu \geq 1$ degrees of freedom, where $\nu$ is fixed and known and the density is given by

$$f(x;\theta) = c_\nu \Big(1 + \frac{(x-\theta)^2}{\nu}\Big)^{-(\nu+1)/2}, \qquad x \in \mathbb{R},$$

with location $\theta \in \mathbb{R}$ and a normalising constant $c_\nu$ free of $\theta$. The log-likelihood is differentiable everywhere, so the recipe seems to apply, but after clearing denominators the likelihood equation

$$\ell'(\theta) = (\nu+1) \sum_{i=1}^{n} \frac{x_i - \theta}{\nu + (x_i - \theta)^2} = 0$$

turns into a polynomial equation of degree $2n-1$ in $\theta$ and for $n \geq 2$ this equation can have several roots. Exercise 5.4 shows that for $n = 2$ there are three roots when the observations are more than $2\sqrt{\nu}$ apart: the mid-point of the interval between the observations, which is then a local *minimum* of the likelihood, and two global maxima of equal likelihood, one close to each of the observations. Thus, the model prefers to fit one data point and to consider the other as an outlier, and in this case the data cannot choose between the two maxima and the MLE will not be unique. The exercise also shows that a root of the likelihood equation can be a minimum and that the checking step in the recipe is not optional.

### No Closed-Form Solution

For larger samples the likelihood equation of the $t$ location model cannot be solved in closed form, and similarly the likelihood equations for many other models, *e.g.* the two-parameter gamma distribution from example 4.9, cannot be solved in closed form. One solution to this problem is to maximise $\ell$ numerically. We will discuss this approach, together with its pitfalls, in lecture 18.

## 5.4 Invariance Under Reparametrisation

Often the quantity we want to estimate is not the parameter itself, but a function of the parameter, *e.g.* a probability, a mean or a median. The following proposition shows that under maximum likelihood such quantities transform together with the parameter.

**Proposition 5.6** (Invariance of the MLE)**.** *Let $g\colon \Theta \to \Psi$ be a bijection and let $\hat{\theta}$ be a maximum likelihood estimator for $\theta$. Then $g(\hat{\theta})$ is a maximum likelihood estimator for the transformed parameter $\psi = g(\theta)$.*

The idea of the proof is that relabelling the parameter does not change the family of distributions: in terms of $\psi$ the likelihood is $\tilde{L}(\psi) = L(g^{-1}(\psi))$, so maximising $\tilde{L}$ over $\Psi$ is the same maximisation as before. Writing this down is exercise 5.5.

*Remark.* If $g$ is not a bijection, then the likelihood of $\psi$ is defined to be the maximal likelihood of the values of $\theta$ which map to this value. While the statement $\hat{\psi} = g(\hat{\theta})$ is still true, we will not require the general case in this module.

As an application, we find the MLE for the median waiting time in the exponential model without any further maximisation.

**Example 5.7.** Let $X_1, \ldots, X_n$ be i.i.d. exponential with rate $\theta > 0$. The *median* waiting time is the value $m$ such that $\mathbb{P}_\theta(X_1 \leq m) = 1/2$. Since $\mathbb{P}_\theta(X_1 \leq m) = 1 - e^{-\theta m}$, this condition is equivalent to $e^{-\theta m} = 1/2$, and taking logarithms gives

$$m = g(\theta) = \frac{\log 2}{\theta}.$$

The map $g$ is strictly decreasing on $(0, \infty)$, with $g(\theta) \to \infty$ as $\theta \to 0$ and $g(\theta) \to 0$ as $\theta \to \infty$, and thus it maps $(0, \infty)$ bijectively onto itself. Since the MLE for the rate is $\hat{\theta} = 1/\bar{X}$ by example 5.2, proposition 5.6 shows that the MLE for the median is

$$\hat{m} = g(\hat{\theta}) = \frac{\log 2}{\hat{\theta}} = (\log 2)\, \bar{X}.$$

Exercise 5.8 shows that this example contains a surprise: $\hat{m}$ is unbiased, despite the MLE for the rate being biased. In general, unbiasedness is lost under nonlinear reparametrisation, and thus there is no analogue of proposition 5.6 for unbiasedness.

Since we have seen in this lecture that the MLE can be biased, non-unique and even non-existent, we need to do more work to understand the method. The arguments for the validity of the MLE can be found in lectures 13, 16 and 17.

*Summary.*

- The maximum likelihood estimator $\hat{\theta}$ is the parameter value which maximises the likelihood (or log-likelihood) of the data.
- For smooth problems, where the maximum is an interior point of the parameter space, the MLE can be found by solving the likelihood equation $\ell'(\theta) = 0$ and verifying that the root is a maximum.
- If this recipe fails, the definition can still be used: for the uniform distribution the maximum sits at the boundary, at $\hat{\theta} = \max_i X_i$, and for the $t$ location model with two observations far apart the MLE is not unique.
- The MLE can be biased.
- The MLE is invariant under reparametrisation: the MLE for $g(\theta)$ is $g(\hat{\theta})$. Unbiasedness is not invariant.

**Exercise 5.1.** Let $X_1, \dots, X_n$ be an i.i.d. sample from the geometric distribution with parameter $p \in (0,1)$, *i.e.* $f(x;p) = (1-p)^{x-1}p$ for $x \in \mathbb{N}$ as in exercise 4.2.

1. Show that the log-likelihood function is given by $\ell(p) = n \log p + \left(\sum_i x_i - n\right) \log(1-p)$.
2. Solve the likelihood equation and show that your solution is a maximum. Deduce that the MLE is $\hat{p} = 1/\bar{X}$.
3. Discuss your results in the light of the method of moments estimator from exercise 4.2.

**Exercise 5.2.** Let $X_1, \dots, X_n$ be i.i.d. with density $f(x;\theta) = \theta^{-2} x\, e^{-x/\theta}$ for $x \geq 0$, where $\theta > 0$. This is the gamma distribution with known shape 2 and scale $\theta$, *i.e.* Gamma$(2, 1/\theta)$ in the rate parametrisation from table A.2.

1. Find the MLE $\hat{\theta}$ for $\theta$.
2. Show that your estimator $\hat{\theta}$ is unbiased.
3. Determine MSE$(\hat{\theta})$ and use this to show that $\hat{\theta}$ is consistent.

**Exercise 5.3.** Let $X_1, \dots, X_n$ be i.i.d. exponential with rate $\theta$, let $\hat{\theta} = 1/\bar{X}$ be the MLE and assume $n \geq 3$.

1. Using the gamma density of $T = X_1 + \cdots + X_n$ as in exercise 2.3, show that $\mathbb{E}_\theta(\hat{\theta}^2) = n^2\theta^2/((n-1)(n-2))$.
2. Deduce that
$$\operatorname{Var}_\theta(\hat{\theta}) = \frac{n^2\theta^2}{(n-1)^2(n-2)}$$
and that the unbiased estimator $\tilde{\theta} = (n-1)\hat{\theta}/n$ from the text has variance $\operatorname{Var}_\theta(\tilde{\theta}) = \theta^2/(n-2)$.

**Exercise 5.4.** Let $x_1 > x_2$ be any two observations from the $t$ location model given in section 5.3. Let $m = (x_1 + x_2)/2$ be the mid-point and $d = (x_1 - x_2)/2$ be half the distance between the two observations.

1. Let $a = x_1 - \theta$ and $b = x_2 - \theta$. Show that the likelihood equation $\ell'(\theta) = 0$ is equivalent to $(a+b)(\nu + ab) = 0$.
2. Show that for $\theta = m + t$ the log-likelihood function is $\ell(m+t) = C - \frac{\nu+1}{2} \log P(t)$, where $C$ does not depend on $t$ and
$$P(t) = (\nu + d^2)^2 + 2(\nu - d^2)\, t^2 + t^4.$$
3. Show that for $d^2 < \nu$ the mid-point $m$ is the unique MLE, but for $d^2 > \nu$ the likelihood equation has three roots $m$, $m \pm \sqrt{d^2 - \nu}$, where $m$ is a local minimum of $\ell$ and the other two are maximum likelihood estimates with equal likelihood.

**Exercise 5.5.** Prove the statement of proposition 5.6: Let $g\colon \Theta \to \Psi$ be a bijection and let the likelihood in the new parametrisation be given by $\tilde{L}(\psi) = L(g^{-1}(\psi))$. Show that $g(\hat{\theta})$ is the maximum of $\tilde{L}$ over $\Psi$.

**Exercise 5.6.** Let $X$ be an observation from the exponential distribution with rate $\theta > 0$, *i.e.* $\mathbb{P}_\theta(X > x) = e^{-\theta x}$ for $x \geq 0$. The following properties of the exponential distribution will be needed later in the module.

1. Show that the exponential distribution has no memory, *i.e.* that for all $s, t \geq 0$ we have

$$\mathbb{P}_\theta(X > s + t | X > s) = \mathbb{P}_\theta(X > t).$$

2. Show that for all $c \geq 0$ we have

$$\mathbb{E}_\theta(X | X > c) = c + \frac{1}{\theta}.$$

   Comment upon your result in the light of the first part of the question.

**Exercise 5.7.** Let $X_1, \ldots, X_n$ be an i.i.d. sample from the Poisson distribution with parameter $\theta > 0$, and assume that $\sum_i x_i \geq 1$.

1. In exercise 4.1 we have seen that the score equals zero at $\bar{x}$. Show that $\bar{X}$ is indeed the MLE for $\theta$.

2. Determine the MLE for the probability $p_0 = \mathbb{P}_\theta(X_1 = 0)$ of observing zero.

3. Using the relation $\mathbb{E}_\theta(z^{X_1}) = e^{\theta(z-1)}$ for $z > 0$, show that your estimator from the previous part is biased, but asymptotically unbiased. Comment on your result in the light of the fact that $\bar{X}$ is unbiased.

**Exercise 5.8.** Let $\hat{m} = (\log 2)\,\bar{X}$ be the MLE for the median waiting time $m = (\log 2)/\theta$ of the exponential distribution from example 5.7. Show that $\hat{m}$ is an unbiased estimator for $m$, and comment on why this does not contradict the bias of the MLE $\hat{\theta} = 1/\bar{X}$ for the rate.

# Lecture 6

# Estimators by Simulation

In the previous two lectures we introduced two estimators for the uniform model: the method of moments estimator $2\bar{X}$ in lecture 4 and the maximum likelihood estimator $\max_i X_i$ in lecture 5. In exercise 4.4 you used algebra to determine which of these two estimators is better. Today we will use the Monte Carlo methods from lecture 3 to answer the same question, and we will learn how to write our own function in R and how to repeat this function in a loop. (Appendix B shows both approaches, but at a slower pace.) The approach we follow here is a good one to get into: before you do any simulation, try to predict what the result will be. Only then can the output of your simulation either confirm or contradict your expectation.

## 6.1 Functions and Loops

The model we consider today is $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$, and to study the estimators for this model using simulation, we will need to repeat the same experiment thousands of times: we have to sample from the model, compute the estimate, and store the result. While we could just type in the commands to do this over and over again in the console, as we did in lecture 3, this approach is not scalable. Instead, we will encapsulate the experiment into a *function* and then use R to repeat this function many times.

To generate samples from the uniform distribution, we can use the function `runif` and to specify the boundaries of the distribution we use `min` and `max`, as explained in appendix B. To create a function, we use the keyword `function`, followed by a list of arguments and then the body of the function enclosed in braces. The return value of a function is the value of the last line of the body. The following function performs a complete experiment for the method of moments estimator: it simulates a fresh sample of size `n` from the model with true parameter `theta`, and then returns the computed value $2\bar{x}$ of the estimate.

```
est_mom <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  2 * mean(sample)
}
set.seed(2026)
est_mom(10, 1)
est_mom(10, 1)
```

```
## [1] 0.8844232
## [1] 0.6339234
```

Defining the function prints nothing; only the two calls do. Each call draws its own sample, so two calls give two different estimates, both scattered around the true value $\theta = 1$: once more the estimator is a random quantity, as seen in lecture 3 for the sample mean.

To repeat the experiment systematically, we can use a `for`-loop: we set aside a vector to store the results using `numeric(5)`, then use a counter `i` which takes the values 1 to 5 (generated by `seq_len(5)`), and then store the estimate for each iteration of the loop in the results vector. (Appendix B explains why we use `seq_len` instead of the shorter notation `1:5`.)

```
set.seed(2026)
estimates <- numeric(5)
for (i in seq_len(5)) {
  estimates[i] <- est_mom(10, 1)
}
estimates
```

```
## [1] 0.8844232 0.6339234 0.5469486 0.9469991 0.8432259
```

Since we reset the seed, the first two values are exactly the two estimates computed above: the loop repeats the same call and keeps the answers. For this everyday pattern of repeating an expression and collecting the results, R provides the shortcut `replicate` which can be used to achieve the same effect in one line.

```
set.seed(2026)
replicate(5, est_mom(10, 1))
```

```
## [1] 0.8844232 0.6339234 0.5469486 0.9469991 0.8432259
```

The five values agree with the loop, as they must. We use `replicate` for the rest of this lecture; the explicit loop returns in lecture 9, where the sample size changes from one run to the next.

## 6.2 Empirical Bias and Mean Squared Error

With `est_mom` in place we can now use the Monte Carlo approach from lecture 3: we generate many estimates and then compute summary statistics. If $\hat{\theta}^{(k)}$ is the estimate using the $k$-th of $N$ simulated samples, then by the law of large numbers (proposition A.14), the average of the deviations $\hat{\theta}^{(k)} - \theta$ is approximately equal to the bias $\mathbb{E}_\theta(\hat{\theta}) - \theta$ from lecture 2, and the average of the squared deviations is approximately equal to the mean squared error $\mathbb{E}_\theta((\hat{\theta} - \theta)^2)$. In R, both summary statistics can be computed in a single line: the empirical bias is given by `mean(estimates) - theta` and the empirical mean squared error is given by `mean((estimates - theta)^2)`.

We now have all the ingredients for a complete Monte Carlo study: a function to perform the experiment, `replicate` to repeat it and two one-liners for the summaries. The box below shows such a study in full, but with the lines of the

R script intentionally shuffled. Before you continue, arrange the lines of this box in the correct order to form an R script which performs a Monte Carlo study with $\theta = 1$, $n = 10$ and 10000 repetitions.

```
estimates <- replicate(10000, est_mom(n, theta))
mean((estimates - theta)^2)    # empirical MSE
theta <- 1
n     <- 10
est_mom <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  2 * mean(sample)
}
mean(estimates) - theta        # empirical bias
set.seed(2026)
```

The rule is that all names must be defined before they are used: the function and the values for `theta` and `n` must come before the `replicate` line, the seed must be set before any random draws are made, and the two summaries need the finished vector `estimates` to be computed, so they come last. There is more than one order which satisfies these conditions, so any of these orders is correct. The reconstructed study is shown below; for comparison the theoretical mean squared error is also shown. Before we run the program, we make a final prediction: from exercise 4.4 we know that $2\bar{X}$ is unbiased and has $\text{MSE}(2\bar{X}) = \theta^2/(3n)$, where $1/30 \approx 0.0333$, so should the two printed summaries be exactly 0 and 0.0333, or just close to these values? How close?

```
est_mom <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  2 * mean(sample)
}
theta <- 1
n     <- 10
set.seed(2026)
estimates <- replicate(10000, est_mom(n, theta))

mean(estimates) - theta        # empirical bias
mean((estimates - theta)^2)    # empirical MSE
theta^2 / (3 * n)              # the theoretical value
```

```
## [1] 0.00226647
## [1] 0.03303851
## [1] 0.03333333
```

The numbers are close to the theory, but not identical, and this is expected: the empirical bias and mean squared error are averages over 10000 random repetitions, so are estimates of the true values, but not identical to the true values. Changing the seed of the random number generator would result in slightly different results. Similarly, if the number of repetitions was increased, the results would get closer to the theoretical values. So, agreement to two decimal places is a good result, and an empirical bias of exactly zero would be a cause for concern (not for celebration).

## 6.3 Comparing Two Estimators

The real use of the Monte Carlo method is to compare estimators in models where the algebra runs out. Before trusting it there, we should watch it succeed on a contest whose outcome we know: by exercises 2.2 and 4.4, the maximum likelihood estimator $\max_i X_i$ of example 5.5 is biased but has smaller mean squared error than the unbiased estimator $2\bar{X}$ for all $n \geq 3$. If the simulation reproduces these known answers, we may trust the same recipe in the practical report and in lecture 18, where no formula is available.

We can write the maximum likelihood estimator as a function, in analogy to `est_mom`. To allow for a fair comparison, we use the same seed `2026` as we used above, to ensure that both estimators use the same 10000 samples. Before you try the code, what should the two printed numbers be close to? From exercise 2.2 we know that the bias of the maximum likelihood estimator is $-\theta/(n+1) = -1/11$ and the mean squared error is $2\theta^2/((n+1)(n+2)) = 2/132 \approx 0.0152$.

```
est_mle <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  max(sample)
}
set.seed(2026)
mle <- replicate(10000, est_mle(n, theta))

mean(mle) - theta              # empirical bias
mean((mle - theta)^2)          # empirical MSE
```

```
## [1] -0.08960997
## [1] 0.01479181
```

Both predictions are correct: the empirical bias $-0.0896$ is close to the theoretical value $-1/11 \approx -0.0909$ and the empirical mean squared error 0.0148 is close to the theoretical value 0.0152, which is less than half of the 0.0330 for $2\bar{X}$ shown above. Thus, in this example, the biased estimator performs better than the unbiased estimator, as shown in exercise 4.4.

The histograms make the comparison visible, and this is the time for the main prediction of the day. Before you run the code, sketch the two histograms you expect, using the theory above for the centres and the widths (the standard deviation of $2\bar{X}$ is $\sqrt{1/30} \approx 0.18$). Which of the two histograms will be more spread out? Where will the two histograms be centred? Will the second histogram be bell-shaped like the first one? The command `par(mfrow = c(2, 1))` can be used to arrange the following two plots as a column, and by using `xlim` for both plots we keep the axes directly comparable, as in lecture 3.

```
par(mfrow = c(2, 1))
hist(estimates, breaks = 40, col = "grey85", main = "method of moments",
     xlab = "estimate", xlim = c(0.4, 1.6))
abline(v = theta, lty = 2, lwd = 3)
hist(mle, breaks = 40, col = "grey85", main = "maximum likelihood",
     xlab = "estimate", xlim = c(0.4, 1.6))
abline(v = theta, lty = 2, lwd = 3)
```

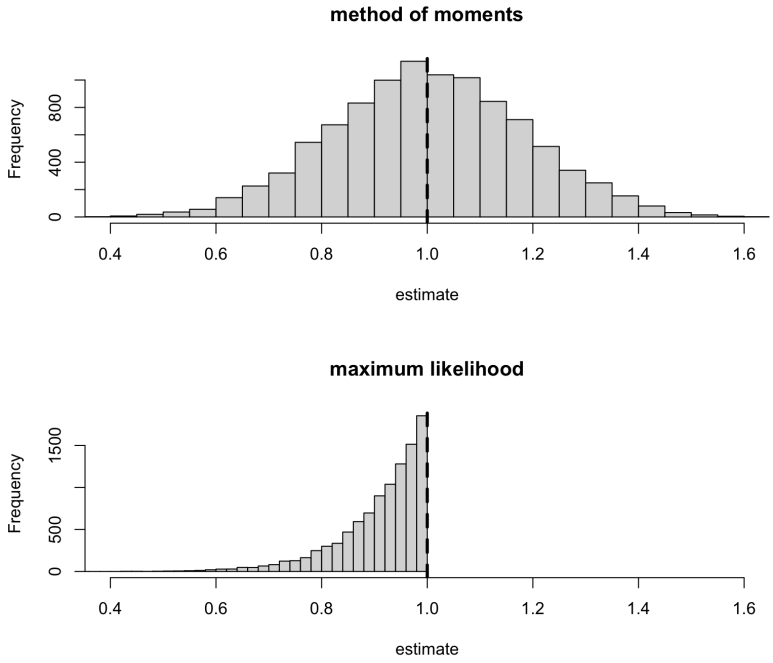



Figure 6.1: Simulated sampling distributions of the two estimators for the uniform parameter, using the same axis: the method of moments estimator $2\bar{X}$ (top) and the maximum likelihood estimator $\max_i X_i$ (bottom). The true value $\theta = 1$ is given by the dashed line.

Figure 6.1 shows that the first two guesses were right, and that the third guess is wrong for most people. The top histogram is a symmetric, bell-shaped histogram, centred around the dashed line: the estimator $2\bar{X}$ is unbiased. The bell-shape is expected, since it is an average and by the central limit theorem (theorem A.15), averages are asymptotically normal. (We will learn more about this result in lecture 12.) The histogram for the maximum likelihood estimator is much narrower, and is shifted to the left of the dashed line. This indicates a bias $-1/11$. The histogram is not bell-shaped at all: no simulated estimates are above 1, since the maximum of a sample from Uniform$(0, \theta)$ can never be above $\theta$ and the estimates are converging to this boundary. This shows that not every estimator has a bell-shaped sampling distribution: the central limit theorem only applies to averages and does not apply to the maximum. We will see this histogram again in lecture 13, where the uniform model will be used as a counterexample to the general theory.

In lecture 9 we will vary the sample size and will see the histograms getting tighter around the truth.

Just type the commands you see in this solution in your own R session, to make sure you understand what they do.

*Summary.*

- Appendix B is the written reference for functions, loops and `replicate`.
- A function created with `function(n, theta) {...}` packages one simulation experiment; a `for`-loop or, more conveniently, `replicate` repeats it and collects the estimates.
- The one-liner `mean(estimates) - theta` gives the empirical bias, and `mean((estimates - theta)^2)` gives the empirical mean squared error; both approximate the true values up to a simulation error that shrinks as the number of repetitions grows.
- For the uniform model with $n = 10$, the simulation reproduced the analytic results of exercise 4.4: the biased estimator $\max_i X_i$ beats the unbiased $2\bar{X}$ in mean squared error, and its sampling distribution is a skewed pile below $\theta$, not a bell.
- Predict before you run: a simulation only teaches when its output can confirm or contradict a stated expectation.

# Lecture 7

# Exponential Families

In lectures 4 and 5 we repeatedly collapsed products of $n$ densities to expressions which summarised the data using only one or two sums. This lecture explains why: almost all families of models considered here have the same algebraic shape, and models of this shape are called *exponential families*. Many of the results in the following weeks are almost automatic for this class.

## 7.1 Recognising the Shape

We start by considering a familiar distribution and then rewrite its probability weights so that the data and the parameter are as far apart as possible. For the Bernoulli distribution with $x \in \{0, 1\}$ we can write

$$\begin{aligned} f(x;\theta) = \theta^x (1-\theta)^{1-x} &= \exp\big(x \log\theta + (1-x)\log(1-\theta)\big) \\ &= \exp\Big(x \log\frac{\theta}{1-\theta} + \log(1-\theta)\Big). \end{aligned}$$

Here we have written the weights as the exponential of the product of one function of $\theta$ and one function of $x$, plus a function of only $\theta$. The Poisson weights and the exponential density can be written in a similar way: for $x \in \mathbb{N}_0$ and $x \geq 0$ respectively we have

$$\frac{\theta^x}{x!} e^{-\theta} = \frac{1}{x!} \exp\big(x \log\theta - \theta\big) \qquad \text{and} \qquad \theta e^{-\theta x} = \exp\big(-\theta x + \log\theta\big).$$

For the Poisson weights an additional factor $1/x!$ appears, which depends on $x$ alone. We will give a name to this structure.

**Definition 7.1.** A parametric model $\big(f(x;\theta) \,\big|\, \theta \in \Theta\big)$ with $\Theta \subseteq \mathbb{R}$ is a *one-parameter exponential family*, if the density or the probability weights can be written as

$$f(x;\theta) = h(x) \exp\big(\eta(\theta)\, T(x) - B(\theta)\big)$$

for all $x$ and all $\theta \in \Theta$, where $h \geq 0$ and $T$ are functions of $x$ alone and $\eta$ and $B$ are functions of $\theta$ alone. The statistic $T(X)$ is called the *natural statistic* and $\eta(\theta)$ is called the *natural parameter* of the family.

Since the exponential is always positive, $f(x;\theta)$ is positive if and only if $h(x)$ is. Thus the support $\{x \mid f(x;\theta) > 0\}$ of the distribution is $\{x \mid h(x) > 0\}$ and does not depend on $\theta$. The representation is not unique, since we can replace $\eta$ by

$c\eta$ and $T$ by $T/c$ for any constant $c \neq 0$. Finally, since the density must integrate (or sum) to one, the function $B$ is determined by the other three functions: we have

$$B(\theta) = \log \int h(x) \exp\big(\eta(\theta)\, T(x)\big)\, \mathrm{d}x. \tag{7.1}$$

These rewritings give the following examples.

**Example 7.2.** The Bernoulli distribution with $\theta \in (0, 1)$ is a one-parameter exponential family with $h(x) = 1$ for $x \in \{0, 1\}$ and $h(x) = 0$ otherwise, natural statistic $T(x) = x$, natural parameter $\eta(\theta) = \log(\theta/(1-\theta))$, the *log-odds* of success, and $B(\theta) = -\log(1-\theta)$.

**Example 7.3.** The Poisson distribution with $\theta > 0$ is a one-parameter exponential family with $h(x) = 1/x!$ for $x \in \mathbb{N}_0$ and $h(x) = 0$ otherwise, natural statistic $T(x) = x$, natural parameter $\eta(\theta) = \log\theta$ and $B(\theta) = \theta$.

**Example 7.4.** The exponential distribution with rate $\theta > 0$ is a one-parameter exponential family with $h(x) = 1$ for $x \geq 0$ and $h(x) = 0$ otherwise, natural statistic $T(x) = x$, natural parameter $\eta(\theta) = -\theta$ and $B(\theta) = -\log\theta$.

The normal distribution is also an exponential family, but since both parameters are unknown, we need to use the two-parameter version of the definition from section 7.2.

*Remark.* The uniform distribution on $[0, \theta]$ is not an exponential family, since the support of the distribution changes with the parameter. In contrast, the support of an exponential family is always fixed. This is the same reason why the likelihood equation was useless in example 5.5. Similarly, the $t$ location model from section 5.3 is not an exponential family, despite the fact that its support is all of $\mathbb{R}$; we do not prove this.

The shape survives the passage from one observation to a sample.

**Lemma 7.5.** *Let $X_1, \ldots, X_n$ be i.i.d. from the one-parameter exponential family in definition 7.1. Then the joint density of the sample is*

$$f(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} h(x_i) \cdot \exp\Big(\eta(\theta) \sum_{i=1}^{n} T(x_i) - nB(\theta)\Big),$$

*which is of the form of definition 7.1 with the same natural parameter $\eta(\theta)$, natural statistic $T_n = \sum_{i=1}^{n} T(X_i)$, the function $\prod_i h(x_i)$ instead of $h$ and $nB(\theta)$ instead of $B(\theta)$.*

*Proof.* Since the observations are independent, the joint density is the product $\prod_{i=1}^{n} h(x_i) \exp\big(\eta(\theta)\, T(x_i) - B(\theta)\big)$ of the marginal densities. Combining the $n$ exponents using the rule $e^a e^b = e^{a+b}$ gives the required form. This completes the proof. □

Read as a function of $\theta$, the joint density is the likelihood, and thus the log-likelihood of an exponential family sample is

$$\ell(\theta) = \sum_{i=1}^{n} \log h(x_i) + \eta(\theta)\, T_n - nB(\theta). \tag{7.2}$$

The first term does not involve $\theta$. Thus, up to a factor which does not depend on $\theta$, the likelihood depends on the data only through the single number $T_n = \sum_i T(x_i)$. This is an extension of example 4.3 to the whole class; in lecture 8 we will see in which sense the value $T_n$ contains all the information about $\theta$.

## 7.2 Several Parameters

The normal distribution where both $\mu$ and $\sigma^2$ are unknown does not fit definition 7.1, since the exponent contains the two products $x\mu/\sigma^2$ and $-x^2/(2\sigma^2)$, which cannot be combined into a single product $\eta(\theta)\,T(x)$. The extension is to allow a sum of $k$ such products in the exponent.

**Definition 7.6.** A parametric model $\big(f(x;\theta)\,\big|\,\theta \in \Theta\big)$ is a *$k$-parameter exponential family*, if the density or the probability weights can be written as

$$f(x;\theta) = h(x)\exp\Big(\sum_{j=1}^{k} \eta_j(\theta)\,T_j(x) - B(\theta)\Big)$$

for all $x$ and all $\theta \in \Theta$, where $h \geq 0$ and $T_1, \ldots, T_k$ are functions of $x$ alone and $\eta_1, \ldots, \eta_k$ and $B$ are functions of $\theta$ alone. The vector $\big(T_1(X), \ldots, T_k(X)\big)$ is the *natural statistic* and $\big(\eta_1(\theta), \ldots, \eta_k(\theta)\big)$ is the *natural parameter* of the family.

The number $k$ counts the products in the exponent, not the components of $\theta$. In most of our examples the two agree, but they do not have to: the normal distribution $N(\theta, \theta^2)$ considered below has the single parameter $\theta > 0$, but its exponent needs two products. An exponential family in which $k$ is larger than the dimension of $\Theta$ is called a *curved exponential family*.

The support of the distribution is again $\{x \mid h(x) > 0\}$ and the proof of lemma 7.5 applies without change: an i.i.d. sample is again a $k$-parameter exponential family, where the natural statistic is the vector of the $k$ sums $\sum_i T_j(X_i)$.

**Example 7.7.** By expanding the square in the exponent of the density of the normal distribution $N(\mu, \sigma^2)$ from table 1.2, we can write

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\Big(-\frac{x^2}{2\sigma^2} + \frac{\mu}{\sigma^2}\,x - \frac{\mu^2}{2\sigma^2}\Big).$$

If only $\mu$ is unknown, the factor $(2\pi\sigma^2)^{-1/2}$ and the term $-x^2/(2\sigma^2)$ contain no unknown quantities and we can define the function $h(x) = (2\pi\sigma^2)^{-1/2}e^{-x^2/(2\sigma^2)}$: we have found a one-parameter exponential family with natural statistic $T(x) = x$, natural parameter $\eta(\mu) = \mu/\sigma^2$ and $B(\mu) = \mu^2/(2\sigma^2)$. If both parameters are unknown, we have to interpret the term $-x^2/(2\sigma^2)$ as a second product and we get a two-parameter exponential family with $h(x) = 1$, natural statistic $(x, x^2)$, natural parameter $\big(\mu/\sigma^2,\, -1/(2\sigma^2)\big)$ and $B(\theta) = \mu^2/(2\sigma^2) + \frac{1}{2}\log(2\pi\sigma^2)$. For a sample, the natural statistic is $\big(\sum_i x_i, \sum_i x_i^2\big)$. Since $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$, for fixed $n$ this pair and the pair $\big(\bar{x}, \sum_i (x_i - \bar{x})^2\big)$ used to derive the maximum likelihood estimator in example 5.4 are one-to-one functions of each other.

The gamma distribution with both parameters unknown is a second example of a two-parameter exponential family. If we write $x^{\alpha-1} = e^{(\alpha-1)\log x}$, then the natural statistic is $(\log x,\, x)$ and the details are worked out in exercise 7.1. For a sample of size $n$, the natural statistic is the pair $\big(\sum_i \log x_i, \sum_i x_i\big)$, which is different from the pair of moments used in example 4.9.

For the completeness statements in lecture 10 it is important that the natural parameter varies in as many directions as there are natural statistics. Mathematically, this can be expressed as the following condition.

**Definition 7.8.** A one-parameter exponential family is of *full rank*, if the set $\eta(\Theta) = \{\eta(\theta) \mid \theta \in \Theta\}$ of natural parameter values contains a non-degenerate open interval. A $k$-parameter exponential family is of *full rank*, if the set $\{(\eta_1(\theta), \ldots, \eta_k(\theta)) \mid \theta \in \Theta\}$ contains a non-empty open subset of $\mathbb{R}^k$.

In the one-parameter case, the condition is almost always satisfied, since the image of an interval $\Theta$ under a continuous, non-constant $\eta$ is an interval with more than one point; all the one-parameter examples above are of full rank. For several parameters, the condition can be more interesting: the natural parameter values of the normal distribution fill the open half-plane $\{(\eta_1, \eta_2) \mid \eta_2 < 0\}$ (example 7.7) and the natural parameter values of the gamma distribution fill an open quadrant (exercise 7.1), but the natural parameter values of $N(\theta, \theta^2)$ with $\theta > 0$, where the standard deviation equals the mean, form a curve in $\mathbb{R}^2$ which does not contain an open set, and thus this curved exponential family is not of full rank (exercise 7.2).

## 7.3 The Natural Parametrisation and the Cumulant Function

We return to one-parameter exponential families. Since the parameter enters the density only via $\eta(\theta)$, it is often convenient to use $\eta$ as the parameter. In this case, the normalising function (7.1) deserves a name, since its first two derivatives are the mean and the variance of the natural statistic.

**Definition 7.9.** Let $h$ and $T$ be as in definition 7.1. Then the *natural parametrisation* of the family is given by

$$f(x;\eta) = h(x)\exp(\eta\, T(x) - K(\eta)), \qquad \eta \in \mathcal{N}.$$

Here the *cumulant function* (also called the *log-partition function*) is given by

$$K(\eta) = \log \int h(x)\, e^{\eta T(x)}\, \mathrm{d}x,$$

with a sum in place of the integral for discrete models. The *natural parameter space* is the set $\mathcal{N} = \{\eta \in \mathbb{R} \mid K(\eta) < \infty\}$ of all $\eta$ for which the integral is finite.

Since $K$ makes $f(x;\eta)$ integrate to one, the formula defines a probability distribution for every $\eta \in \mathcal{N}$, whether or not $\eta$ is of the form $\eta(\theta)$; comparing with (7.1) shows that $\eta(\theta) \in \mathcal{N}$ and $B(\theta) = K(\eta(\theta))$, so that the original family sits inside this larger one. For the three examples of section 7.1 we find $K(\eta) = \log(1 + e^\eta)$ with $\mathcal{N} = \mathbb{R}$ for the Bernoulli distribution, $K(\eta) = e^\eta$ with $\mathcal{N} = \mathbb{R}$ for the Poisson distribution and $K(\eta) = -\log(-\eta)$ with $\mathcal{N} = (-\infty, 0)$ for the exponential distribution. In all three cases $\mathcal{N}$ is an open interval and $\theta \mapsto \eta(\theta)$ is a bijection onto $\mathcal{N}$, making the natural parametrisation a relabelling in the sense of proposition 5.6.

The first two derivatives of $K$ give the mean and the variance of $T(X)$, and the name of $K$ comes from a related object. The *moment generating function* of a random variable $Y$ is the function $s \mapsto \mathbb{E}(e^{sY})$, defined for all $s$ for which the expectation is finite. Its logarithm $s \mapsto \log \mathbb{E}(e^{sY})$ is called the *cumulant generating function* of $Y$, and the derivatives of this function at $s = 0$ are called

the *cumulants* of $Y$. The first two cumulants are the mean and the variance of $Y$. For a random variable $Y$ with values in $\mathbb{N}_0$ the closely related *probability generating function* is given by

$$G_Y(z) = \mathbb{E}(z^Y) = \sum_{y=0}^{\infty} \mathbb{P}(Y = y)\, z^y, \tag{7.3}$$

defined for all $z \geq 0$ for which the sum is finite. Substituting $z = e^s$ shows that $s \mapsto G_Y(e^s)$ is the moment generating function of $Y$. In exercise 7.6 we show that $s \mapsto K(\eta + s) - K(\eta)$ is the cumulant generating function of $T(X)$, which explains the name of $K$.

**Lemma 7.10.** *Let $\eta$ be an interior point of the natural parameter space $\mathcal{N}$. Then, in the natural parametrisation,*

$$\mathbb{E}_\eta(T(X)) = K'(\eta) \qquad \textit{and} \qquad \mathrm{Var}_\eta(T(X)) = K''(\eta).$$

*Proof.* Set $Z(\eta) = \int h(x) e^{\eta T(x)}\, \mathrm{d}x$, so that $K = \log Z$. Since the density $f(x;\theta) = h(x) \exp(\eta(\theta)\, T(x) - B(\theta))$ integrates to one, the function $h \geq 0$ is not identically zero. Together with $e^{\eta T(x)} > 0$ this gives $Z(\eta) > 0$, and thus we may divide by $Z(\eta)$ below. Since $\eta$ is in the interior of $\mathcal{N}$, we can take derivatives inside the integral as many times as required; this is a result from analysis which we use without proof. Taking derivatives once, we get

$$Z'(\eta) = \int T(x)\, h(x)\, e^{\eta T(x)}\, \mathrm{d}x.$$

Dividing by $Z(\eta) = e^{K(\eta)}$ changes the integrand to $T(x)$ times the density $f(x;\eta)$ and thus we get

$$K'(\eta) = \frac{Z'(\eta)}{Z(\eta)} = \int T(x)\, h(x)\, e^{\eta T(x) - K(\eta)}\, \mathrm{d}x = \mathbb{E}_\eta(T(X)).$$

Taking derivatives again, we find

$$K''(\eta) = \frac{Z''(\eta)}{Z(\eta)} - \Big(\frac{Z'(\eta)}{Z(\eta)}\Big)^2,$$

and using the same argument as above, with $T(x)^2$ in the integrand, we find $Z''(\eta)/Z(\eta) = \mathbb{E}_\eta(T(X)^2)$. Thus we have $K''(\eta) = \mathbb{E}_\eta(T(X)^2) - \mathbb{E}_\eta(T(X))^2 = \mathrm{Var}_\eta(T(X))$ and the proof is complete. □

The lemma allows us to compute moments by differentiation: for the Poisson distribution $K(\eta) = e^\eta$ we find $K'(\eta) = K''(\eta) = e^\eta = \theta$, the mean and variance from table A.1, and in exercise 7.5 we consider the exponential and normal distributions. For $T_n$, being a sum of $n$ i.i.d. copies of $T(X)$, we can use the lemma to find $\mathbb{E}_\eta(T_n) = nK'(\eta)$ and $\mathrm{Var}_\eta(T_n) = nK''(\eta)$.

## 7.4 What the Structure Buys

We conclude this lecture by collecting what the exponential family shape tells us about the questions of lectures 4 and 5, and by indicating where the shape will be used again.

The first consequence concerns the likelihood equation. By lemma 7.5, written in the natural parametrisation, the log-likelihood of a sample is

$$\ell(\eta) = \sum_{i=1}^{n} \log h(x_i) + \eta\, T_n - nK(\eta),$$

and thus we have $\ell'(\eta) = T_n - nK'(\eta)$ and, by lemma 7.10, $\ell''(\eta) = -nK''(\eta) = -n \operatorname{Var}_\eta(T(X))$. If $T(X)$ is not constant, this variance is positive and $\ell$ is strictly concave on the interior of $\mathcal{N}$. Since $\ell'$ is then strictly decreasing, the likelihood equation $\ell'(\eta) = 0$ has at most one root, and if there is a root, it is the maximum likelihood estimator for $\eta$. When $\theta \mapsto \eta(\theta)$ is a bijection, proposition 5.6 transfers this estimator back to $\theta$. The argument shows uniqueness, but not existence: for a Bernoulli sample with no successes we have $T_n = 0$, and the equation $K'(\eta) = e^\eta/(1+e^\eta) = 0$ has no solution. This is the case $s = 0$ from the remark after example 5.3.

The second consequence concerns smoothness. Many of the results later in the module are proved by differentiating with respect to the parameter, and they require the support of the distribution not to depend on $\theta$, the density to be smooth in $\theta$ and derivatives to be taken inside integrals. For an exponential family the support is fixed by definition, the density is smooth in $\theta$ whenever $\eta$ and $B$ are, and the argument in the proof of lemma 7.10 allows differentiation under the integral sign whenever $\eta(\theta)$ lies in the interior of $\mathcal{N}$. Similarly, if $\eta$ is one-to-one and $T(X)$ is not constant, then different values of $\theta$ give different distributions, and the parameter is identifiable in the sense of definition 4.10. Under these mild conditions on $\eta$, the exponential families are the standard examples for the theory in the rest of the module.

Finally, the observation that the likelihood depends on the data only through $T_n$ and $n$ suggests that, once $T_n$ is known, the individual observations carry no further information about $\theta$. In lecture 8 we make this idea precise, and lemma 7.5 will do most of the work there. From then on, the natural statistic $T_n$ will be the starting point whenever we look for good estimators, and again when we update our knowledge about $\theta$ using the Bayesian approach in the second half of the module.

*Summary.*

- A one-parameter exponential family has density of the form $f(x;\theta) = h(x)\exp(\eta(\theta)T(x) - B(\theta))$, where the natural statistic is $T$, the natural parameter is $\eta$ and the support of the distribution does not depend on $\theta$. Examples of exponential families include the Bernoulli, Poisson, exponential, normal and gamma distributions. The uniform distribution is not an exponential family.
- An i.i.d. sample from an exponential family is again an exponential family. In this case, the natural statistic is $T_n = \sum_i T(X_i)$ and the likelihood depends on the data only through $T_n$.
- A $k$-parameter exponential family is of full rank, if the set of natural parameter values contains a non-empty open subset of $\mathbb{R}^k$. This condition is required for completeness in lecture 10. The curved exponential family $N(\theta, \theta^2)$ is not of full rank.
- In the natural parametrisation $f(x;\eta) = h(x)\exp(\eta T(x) - K(\eta))$ the cumulant function $K$ can be used to find the mean and variance of the natural statistic: we have $\mathbb{E}_\eta(T(X)) = K'(\eta)$ and $\mathrm{Var}_\eta(T(X)) = K''(\eta)$.
- If $T(X)$ is not constant, the log-likelihood is strictly concave in the natural parameter, and thus the likelihood equation has at most one root. A root need not exist.
- Under mild conditions on $\eta$, exponential families satisfy the smoothness conditions of the later theory.

**Exercise 7.1.** Let $X$ be gamma distributed, $X \sim \mathrm{Gamma}(\alpha, \beta)$ with shape $\alpha > 0$ and rate $\beta > 0$, both unknown, *i.e.* with density $f(x;\alpha,\beta) = \beta^\alpha x^{\alpha-1} e^{-\beta x}/\Gamma(\alpha)$ for $x > 0$, as listed in table A.2.

1. Show that this model is a two-parameter exponential family, as given in definition 7.6, with $h(x) = 1$ for $x > 0$ and $h(x) = 0$ otherwise, natural statistic $(\log x,\ x)$, natural parameter $(\alpha - 1,\ -\beta)$, and function $B(\alpha,\beta) = \log\Gamma(\alpha) - \alpha\log\beta$.

2. Show that the set of natural parameter values is the open quadrant

$$\{(\eta_1,\eta_2) \mid \eta_1 > -1,\ \eta_2 < 0\}$$

   and thus the family has full rank as given in definition 7.8.

3. Assume that the shape $\alpha$ is known, as in exercise 5.2. Show that then the factor $x^{\alpha-1}$ can be incorporated into the function $h$ to yield a one-parameter exponential family with natural statistic $T(x) = x$ and natural parameter $-\beta$.

**Exercise 7.2.** Consider the normal distribution $N(\theta, \theta^2)$ with mean and standard deviation both equal to $\theta > 0$.

1. Using example 7.7, show that this model is a curved two-parameter exponential family with natural statistic and natural parameter given by

$$T(x) = (x, x^2) \qquad \text{and} \qquad \eta(\theta) = (1/\theta,\ -1/(2\theta^2)).$$

2. Show that as $\theta$ varies, the natural parameter values follow the curve $\eta_2 = -\eta_1^2/2$ with $\eta_1 > 0$. Conclude that this family is not of full rank in the sense of definition 7.8.

**Exercise 7.3.** Let $X$ be geometrically distributed with parameter $p \in (0,1)$, *i.e.* $f(x;p) = (1-p)^{x-1}p$ for $x \in \mathbb{N}$, as in exercises 4.2 and 5.1.

1. Show that this distribution is a one-parameter exponential family and identify the functions $h$, $T$, $\eta$ and $B$. What is the natural statistic of an i.i.d. sample $X_1, \ldots, X_n$?

2. Determine the natural parameter space $\mathcal{N}$ and the cumulant function $K$. Check that $B(p) = K\big(\eta(p)\big)$. Is this family of full rank?

3. Using lemma 7.10, determine the mean and variance of $X$ and compare the result with table A.1.

**Exercise 7.4.** Let $X$ be binomially distributed with a known number $m$ of trials and unknown success probability $\theta \in (0,1)$, *i.e.* $f(x;\theta) = \binom{m}{x}\theta^x(1-\theta)^{m-x}$ for $x \in \{0,1,\ldots,m\}$.

1. Show that this distribution is a one-parameter exponential family, with the same natural parameter as the Bernoulli distribution from example 7.2, and identify the functions $h$, $T$ and $B$.

2. Determine the natural parameter space and the cumulant function.

3. Using lemma 7.10, determine the mean and variance of $X$, and the mean and variance of the natural statistic $T_n$ for an i.i.d. sample of size $n$.

**Exercise 7.5.**

1. For the exponential distribution with rate $\theta$, use the cumulant function $K(\eta) = -\log(-\eta)$ from the text to find the mean and variance of an observation.

2. For the normal distribution with known variance $\sigma^2$ and unknown mean $\mu$, where $h$, $T$ and $\eta$ are as in example 7.7, use the method of completing the square to show that $K(\eta) = \sigma^2\eta^2/2$ with $\mathcal{N} = \mathbb{R}$ and use this to derive the mean and variance of an observation. Check that your result satisfies $B(\mu) = K\big(\eta(\mu)\big)$.

**Exercise 7.6.** Let $X$ be a random variable with density

$$f(x;\eta) = h(x)\exp\big(\eta T(x) - K(\eta)\big)$$

given by definition 7.9, and let $s$ be such that $\eta + s \in \mathcal{N}$.

1. Show that $\mathbb{E}_\eta\big(e^{sT(X)}\big) = \exp\big(K(\eta+s) - K(\eta)\big)$, *i.e.* that $C(s) = K(\eta + s) - K(\eta)$ is the cumulant generating function of $T(X)$.

2. Let $\eta$ be an interior point of $\mathcal{N}$. Then, since $C$ is differentiable at $s = 0$, we can find $C'(0)$ and $C''(0)$ from the first part of this exercise. Use these results to give an alternative proof of lemma 7.10.

3. Using the first part of this exercise, show that for the Poisson distribution with parameter $\theta$ the probability generating function (7.3) is given by $\mathbb{E}_\theta(z^X) = e^{\theta(z-1)}$ for all $z > 0$. This is the same identity as in exercise 5.7.

**Exercise 7.7.** Let $X_1, \ldots, X_n$ be an i.i.d. sample from a one-parameter exponential family in its natural parametrisation, with $\mathcal{N}$ an open interval and $\mathrm{Var}_\eta(T(X_1)) > 0$ for all $\eta \in \mathcal{N}$.

1. Show that the likelihood equation $\ell'(\eta) = 0$ from lecture 5 can be written as
$$\frac{1}{n}\sum_{i=1}^{n} T(x_i) = \mathbb{E}_\eta(T(X_1)),$$
*i.e.* the maximum likelihood estimator is the parameter value under which the population mean of the natural statistic equals its sample mean.

2. Show that every root of the likelihood equation is the unique maximum likelihood estimator for $\eta$.

3. Apply the equation from the first part to the Poisson distribution and to the Bernoulli distribution, and use proposition 5.6 to recover the estimators found in exercise 5.7 and example 5.3.

4. Explain how the first part of this exercise accounts for the coincidences between the method of moments and maximum likelihood which we noted in lecture 4, and why no such coincidence is to be expected for the gamma distribution of exercise 7.1.

# Lecture 8

# Sufficiency

In example 4.2 we have observed ten coin tosses and got seven heads. Writing $\theta$ for the heads probability, the likelihood in this example is $L(\theta) = \theta^7(1-\theta)^3$. This function depends on the data only via the number 7 of heads, but not on the order of heads and tails, *i.e.* for learning about $\theta$ we can replace the full record of observations by this single number. This lecture makes the idea precise: a *sufficient statistic* contains all information about the parameter. The factorisation theorem helps to find sufficient statistics, and the concept of sufficiency forms the basis of improvements of estimators in lecture 10.

In this lecture we write $X = (X_1, \ldots, X_n)$ for the data, $x = (x_1, \ldots, x_n)$ for an observed value, and

$$f(x;\theta) = \prod_{i=1}^{n} f(x_i;\theta)$$

for the joint density or joint probability weights of the sample; the vector argument distinguishes it from the density of a single observation. As a function of $\theta$ for fixed data, this is the likelihood $L(\theta)$ from lecture 4.

## 8.1 Sufficient Statistics

A statistic $T = T(X_1, \ldots, X_n)$ in the sense of definition 1.3 reduces the data to a smaller summary, and usually some information is lost in the process. The following definition expresses that no information about $\theta$ is lost.

**Definition 8.1.** A statistic $T = T(X_1, \ldots, X_n)$ is *sufficient* for $\theta$, if the conditional distribution of $(X_1, \ldots, X_n)$ given the value $T = t$ does not depend on $\theta$, for all values $t$ of $T$.

If $T$ is sufficient, anyone who knows the value $t$ of $T$ can generate a new data set with the same distribution as the original data set, by sampling from the conditional distribution of $X$ given $T = t$. Since this distribution does not depend on $\theta$, all information about $\theta$ that we could obtain by studying the data set can now be obtained by studying $t$ alone.

**Example 8.2.** Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability $p$ and let $T = \sum_{i=1}^{n} X_i$ be the number of successes. We know that $T \sim \text{Binomial}(n, p)$, as a sum of $n$ Bernoulli variables. Let $t \in \{0, 1, \ldots, n\}$ and let $x \in \{0, 1\}^n$ be a

data vector with $\sum_i x_i = t$. Since the event $\{X = x\}$ is contained in the event $\{T = t\}$, we have

$$\mathbb{P}_p(X = x|T = t) = \frac{\mathbb{P}_p(X = x)}{\mathbb{P}_p(T = t)} = \frac{p^t(1-p)^{n-t}}{\binom{n}{t}p^t(1-p)^{n-t}} = \frac{1}{\binom{n}{t}}.$$

If $\sum_i x_i \neq t$, the events $\{X = x\}$ and $\{T = t\}$ are disjoint and the conditional probability is zero. In both cases the value does not depend on $p$, and thus $T$ is sufficient for $p$. Given the number of successes, each of the $\binom{n}{t}$ possible arrangements of successes and failures is equally likely, whatever the value of $p$: the positions of the successes do not contain any information about the coin.

Two simple observations follow from the definition. First, the whole sample $T = (X_1, \ldots, X_n)$ is always sufficient, since given the whole sample there is no randomness left. Secondly, if $\varphi$ is a one-to-one function, the events $\{\varphi(T) = \varphi(t)\}$ and $\{T = t\}$ coincide, and thus $\varphi(T)$ is sufficient whenever $T$ is: a one-to-one function of a sufficient statistic is sufficient again. For the coin tosses the sample mean $\bar{X} = T/n$ is thus sufficient too.

## 8.2 The Factorisation Theorem

Directly verifying the definition of sufficiency requires knowledge of the distribution of $T$ and involves a conditional probability which, for most models, is tedious to compute. The following proposition replaces the computation with a condition on the density or weights.

**Proposition 8.3** (Fisher–Neyman factorisation theorem)**.** *The statistic $T$ is sufficient for $\theta$, if and only if there are functions $h$ and $g$ such that*

$$f(x;\theta) = h(x)\, g\big(T(x);\theta\big) \tag{8.1}$$

*for all data vectors $x$ and all $\theta \in \Theta$.*

The function $h$ may not depend on $\theta$ at all, and the function $g$ may depend on the data only through $T(x)$: the parameter and the data interact only through $T$. Nothing in the statement requires $T$ or $\theta$ to be one-dimensional. We give a proof for discrete data only.

*Proof for discrete data.* Since we have discrete observations, the joint probability weight $f(x;\theta) = \mathbb{P}_\theta(X = x)$ is the probability of observing the data vector $x$.

Assume first that $T$ is sufficient and let $x$ be a data vector with $t = T(x)$. Since the event $\{X = x\}$ is contained in the event $\{T = t\}$, we have

$$\mathbb{P}_\theta(X = x) = \mathbb{P}_\theta(X = x,\, T = t) = \mathbb{P}_\theta(X = x|T = t)\, \mathbb{P}_\theta(T = t)$$

whenever $\mathbb{P}_\theta(T = t) > 0$. Using the sufficiency of the statistic, the conditional probability does not depend on $\theta$ and thus is a function $h(x)$ of the data. The remaining factor is a function $g(t;\theta)$ of $t$ and $\theta$. If $\mathbb{P}_\theta(T = t) = 0$ for the given $\theta$, both sides in (8.1) equal zero since $g(t;\theta) = 0$. For the data vectors $x$ with $\mathbb{P}_\theta(T = T(x)) = 0$ for every $\theta$, which are never observed, we define $h(x) = 0$. Thus (8.1) holds for all $x$ and $\theta$.

Conversely, assume that (8.1) holds and let $t$ be a value of $T$ and $\theta$ a parameter value with $\mathbb{P}_\theta(T = t) > 0$. Summing the factorisations for all data vectors $y$ with $T(y) = t$, we find

$$\mathbb{P}_\theta(T = t) = \sum_{y\colon T(y)=t} f(y;\theta) = \sum_{y\colon T(y)=t} h(y)\, g(t;\theta) = H(t)\, g(t;\theta),$$

where $H(t)$ is the sum of the $h(y)$, which does not depend on $\theta$. Since the left-hand side is positive, we have $g(t;\theta) \neq 0$. For data $x$ with $T(x) = t$ we get

$$\mathbb{P}_\theta(X = x | T = t) = \frac{\mathbb{P}_\theta(X = x)}{\mathbb{P}_\theta(T = t)} = \frac{h(x)\, g(t;\theta)}{H(t)\, g(t;\theta)} = \frac{h(x)}{H(t)},$$

and for $x$ with $T(x) \neq t$ the conditional probability is zero. Thus, in both cases the value does not depend on $\theta$ and $T$ is sufficient. This completes the proof. □

The proof for the continuous case is more technical, and we omit it here. As a first step, one needs to define what it means to condition the distribution of $X$ on $T = t$, since we typically have $\mathbb{P}(T = t) = 0$ in the continuous case. To avoid these complications, we will use (8.1) as the definition of sufficiency for the continuous case from now on.

When the support depends on $\theta$, the density has to be written down for all $x$, including the values where it is zero, since this is where the parameter and the data may interact. To write such densities compactly we use the *indicator function* of a condition, $\mathbf{1}_{\{A\}}$, which equals 1 if the condition $A$ holds and 0 otherwise. The uniform distribution is the standard example.

**Example 8.4.** Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ be i.i.d. with $\theta > 0$, as in example 1.2. The density of a single observation, written for all $x_i \in \mathbb{R}$, is $f(x_i;\theta) = \theta^{-1}\mathbf{1}_{\{0 \leq x_i \leq \theta\}}$. A product of indicators is the indicator of the intersection of the conditions, and all $n$ conditions $0 \leq x_i \leq \theta$ hold if and only if the smallest observation is at least 0 and the largest is at most $\theta$. Thus the joint density is given by

$$f(x;\theta) = \theta^{-n} \prod_{i=1}^{n} \mathbf{1}_{\{0 \leq x_i \leq \theta\}} = \mathbf{1}_{\{\min_i x_i \geq 0\}}\, \theta^{-n}\, \mathbf{1}_{\{\max_i x_i \leq \theta\}}.$$

If we set $h(x) = \mathbf{1}_{\{\min_i x_i \geq 0\}}$ and $g(t;\theta) = \theta^{-n}\mathbf{1}_{\{t \leq \theta\}}$ we have

$$f(x;\theta) = h(x) g\big(\max_i x_i;\, \theta\big)$$

and by proposition 8.3 the sample maximum $X_{(n)} = \max_i X_i$ is sufficient for $\theta$. It is a common mistake to try to write the joint density simply as $\theta^{-n}$ and to conclude that no statistic is needed: the indicator is where the data and the parameter interact.

The factorisation has a consequence for maximum likelihood estimators. For fixed data the likelihood is $L(\theta) = h(x)\, g\big(T(x);\theta\big)$, and the factor $h(x)$ does not affect where the maximum over $\theta$ is attained: the maximum likelihood estimator, whenever it is unique, is a function of any sufficient statistic. The examples of lecture 5 confirm this: the Bernoulli MLE is $T/n$ with the sufficient statistic $T$ of example 8.2, and the uniform MLE is the sample maximum, which example 8.4 shows to be sufficient.

## 8.3 Sufficiency in Exponential Families

For the exponential families from lecture 7, no work is needed to find a sufficient statistic, since the densities are already given in factorised form.

**Corollary 8.5.** *Let $X_1, \dots, X_n$ be an i.i.d. sample from a one-parameter exponential family as given in definition 7.1 and let $T$ be the natural statistic. Then $T_n = \sum_{i=1}^n T(X_i)$ is a sufficient statistic for $\theta$.*

*For a $k$-parameter exponential family as given in definition 7.6, the vector $\big(\sum_i T_1(X_i), \dots, \sum_i T_k(X_i)\big)$ of summed natural statistics is sufficient for the parameter vector.*

*Proof.* From lemma 7.5 we know that the joint density of the sample is

$$f(x;\theta) = \Big(\prod_{i=1}^n h(x_i)\Big) \exp\Big(\eta(\theta) \sum_{i=1}^n T(x_i) - nB(\theta)\Big).$$

Thus, we can write the joint density in the form (8.1) with the product $\prod_i h(x_i)$ in the role of $h(x)$ and with $g(t;\theta) = \exp\big(\eta(\theta)t - nB(\theta)\big)$. Consequently, $T_n$ is sufficient by proposition 8.3. A very similar argument gives the proof for $k$-parameter exponential families. □

**Example 8.6.** Let $X_1, \dots, X_n$ be i.i.d. Poisson with parameter $\theta > 0$. The joint probability weights are given by

$$f(x;\theta) = \prod_{i=1}^n \frac{\theta^{x_i}}{x_i!}\, e^{-\theta} = \theta^{\sum_i x_i}\, e^{-n\theta} \cdot \frac{1}{\prod_i x_i!},$$

and thus the sum $T = \sum_i X_i$ is sufficient for $\theta$, with $g(t;\theta) = \theta^t e^{-n\theta}$ and $h(x) = 1/\prod_i x_i!$.

**Example 8.7.** Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ be i.i.d. with both parameters unknown. Example 7.7 identifies the natural statistics as $x$ and $x^2$, and thus by corollary 8.5 the pair $T = \big(\sum_i X_i, \sum_i X_i^2\big)$ is sufficient for $(\mu, \sigma^2)$. Since $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$, the pair $(\bar{X}, S^2)$ maps to $T$ one-to-one and thus is also sufficient. This is the pair of statistics on which the maximum likelihood estimator from example 5.4 is based; the factorisation is worked out in exercise 8.4.

## 8.4 The Sample Maximum and Minimum

Example 8.4 shows that a sufficient statistic need not be a sum. To find the bias or the mean squared error of the sample maximum we need its distribution, which the following result gives for the maximum and minimum of an i.i.d. sample.

**Proposition 8.8.** *Let $X_1, \dots, X_n$ be i.i.d. with distribution function $F$ and density $f$. Furthermore, let $X_{(n)} = \max_i X_i$ and $X_{(1)} = \min_i X_i$. Then the random variables $X_{(n)}$ and $X_{(1)}$ have densities*

$$f_{(n)}(y) = n\, F(y)^{n-1} f(y) \qquad \text{and} \qquad f_{(1)}(y) = n\,\big(1 - F(y)\big)^{n-1} f(y).$$

*Proof.* The maximum is at most $y$, if and only if all observations are at most $y$. Using independence we get

$$\mathbb{P}(X_{(n)} \leq y) = \mathbb{P}(X_1 \leq y, \ldots, X_n \leq y) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq y) = F(y)^n.$$

Using the chain rule for differentiation with respect to $y$, we find that the maximum has density $nF(y)^{n-1}f(y)$. Similarly, the minimum is strictly greater than $y$, if and only if all observations are strictly greater than $y$. Thus we get

$$\mathbb{P}(X_{(1)} \leq y) = 1 - \mathbb{P}(X_1 > y, \ldots, X_n > y) = 1 - \big(1 - F(y)\big)^n.$$

Using the chain rule for differentiation we find $n\big(1 - F(y)\big)^{n-1} f(y)$. This completes the proof. □

**Example 8.9.** Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ be i.i.d., as in example 8.4. The distribution function is $F(y) = y/\theta$ for $0 \leq y \leq \theta$, and the density is $f(y) = 1/\theta$ on $[0, \theta]$. By proposition 8.8 the sample maximum has density

$$f_{(n)}(y) = n\Big(\frac{y}{\theta}\Big)^{n-1} \frac{1}{\theta} = \frac{n\, y^{n-1}}{\theta^n}, \qquad 0 \leq y \leq \theta,$$

and $f_{(n)}(y) = 0$ outside this interval. This is the density supplied without proof in exercise 2.2, from which we computed $\mathbb{E}_\theta(X_{(n)}) = n\theta/(n+1)$ there. The support of the maximum moves with $\theta$, as the support of a single observation does.

## 8.5 Minimal Sufficiency and Completeness

As we have seen in section 8.1, the whole sample is always sufficient. For an i.i.d. sample, the vector of ordered observations is also sufficient, since the joint density does not change when the $x_i$ are interchanged. The aim of this section is to characterise the statistic which allows us to reduce the data as much as possible, while still retaining all the information.

**Definition 8.10.** A sufficient statistic $T$ is *minimal sufficient* for $\theta$, if for every sufficient statistic $S$ there is a function $\varphi$ such that $T = \varphi(S)$.

A minimal sufficient statistic can thus be computed from every other sufficient statistic. The definition is awkward to check directly, because it refers to all sufficient statistics at once; the following criterion instead compares the likelihood functions of two data sets. In the criterion we write

$$\mathcal{X} = \{\, x \mid f(x;\theta) > 0 \text{ for some } \theta \in \Theta \,\}$$

for the set of data vectors which can be observed under some parameter value, for the Poisson distribution the vectors of non-negative integers.

**Proposition 8.11.** *Let $T$ be a statistic such that for all $x, y \in \mathcal{X}$ the following two statements are equivalent:*

1. *there is a constant $c(x,y) > 0$ such that $f(x;\theta) = c(x,y)\, f(y;\theta)$ for all $\theta \in \Theta$;*

2. $T(x) = T(y)$.

*Then $T$ is minimal sufficient for $\theta$.*

When both $f(x;\theta)$ and $f(y;\theta)$ are positive for all $\theta$, the first statement says that the ratio $f(x;\theta)/f(y;\theta)$ does not depend on $\theta$. We only sketch the proof. For sufficiency, we choose for every value $t$ of $T$ a data vector $x_t \in \mathcal{X}$ with $T(x_t) = t$; the first statement then gives the factorisation (8.1) with $g(t;\theta) = f(x_t;\theta)$ and $h(x) = c(x, x_t)$. For minimality, let $S$ be sufficient and $x, y \in \mathcal{X}$ with $S(x) = S(y)$. The factorisation for $S$ shows that $f(x;\theta)$ and $f(y;\theta)$ are proportional, and the criterion gives $T(x) = T(y)$: the value of $T$ is determined by the value of $S$.

The criterion states that $T(x)$ determines the likelihood function of the data up to a constant factor, and that two data sets with different values of $T$ have likelihood functions which are not proportional. We apply the criterion once, to the Poisson sum. The two-parameter normal case is worked out in exercise 8.4.

**Example 8.12.** Let $X_1, \ldots, X_n$ be i.i.d. Poisson with parameter $\theta > 0$, and let $T = \sum_i X_i$. For two vectors $x$ and $y$ of non-negative integers, both probability weights are positive and the joint probability weights of example 8.6 give

$$\frac{f(x;\theta)}{f(y;\theta)} = \frac{\theta^{\sum_i x_i} e^{-n\theta} / \prod_i x_i!}{\theta^{\sum_i y_i} e^{-n\theta} / \prod_i y_i!} = \theta^{\sum_i x_i - \sum_i y_i} \frac{\prod_i y_i!}{\prod_i x_i!}.$$

If $\sum_i x_i = \sum_i y_i$, the power of $\theta$ disappears and the ratio is a positive constant. If the sums differ by $k \neq 0$, the ratio contains the factor $\theta^k$, which is not constant as $\theta$ ranges over $(0, \infty)$. Thus the ratio is free of $\theta$ if and only if $\sum_i x_i = \sum_i y_i$, and $T$ is minimal sufficient by proposition 8.11.

For the results of lecture 10 we will need one more property of a statistic, concerning functions of the statistic which are unbiased estimators for zero.

**Definition 8.13.** A statistic $T$ is *complete*, if for every function $w$ such that $w(T)$ has finite mean under $\mathbb{P}_\theta$ for all $\theta \in \Theta$, the condition $\mathbb{E}_\theta(w(T)) = 0$ for all $\theta \in \Theta$ implies $\mathbb{P}_\theta(w(T) = 0) = 1$ for all $\theta \in \Theta$.

This means that a function of a complete statistic which has expectation zero for all parameter values is itself zero with probability one. The condition is violated for statistics which do not reduce the data enough: in exercise 8.6 we will see that the whole Bernoulli sample is sufficient, but not complete, whereas the total $\sum_i X_i$ is complete by proposition 10.5 in lecture 10. Completeness is a separate property from minimal sufficiency: a minimal sufficient statistic need not be complete. The property matters because two unbiased estimators for the same quantity which are both functions of a complete statistic agree with probability one. In lecture 10 we will use this uniqueness to turn the Rao–Blackwell improvement into an optimality result.

*Summary.*

- A statistic $T$ is sufficient for $\theta$, if the conditional distribution of the data given $T$ does not depend on $\theta$: the data can be reduced to $T$ without losing information about $\theta$.
- The factorisation theorem states that $T$ is sufficient if and only if $f(x;\theta) = h(x)\, g(T(x);\theta)$. When the support depends on $\theta$, its indicator is part of the density and belongs to $g$. When the MLE is unique, it is a function of any sufficient statistic.
- For exponential families, the summed natural statistic $\sum_i T(X_i)$ is sufficient. This includes the Bernoulli total, the Poisson sum and the normal pair $(\sum_i X_i, \sum_i X_i^2)$.
- The densities of the sample maximum and minimum are $nF^{n-1}f$ and $n(1-F)^{n-1}f$, obtained by differentiating $F^n$ and $1-(1-F)^n$, respectively.
- A minimal sufficient statistic is a function of any sufficient statistic. A statistic $T$ is minimal sufficient, if $T(x) = T(y)$ holds exactly when the likelihood functions of $x$ and $y$ are proportional. A complete statistic is a statistic such that every function of it with expectation zero for all $\theta$ is zero with probability one. In lecture 10 we will use this property to identify the unique best unbiased estimator.

**Exercise 8.1.** Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability $\theta \in (0,1)$, where $n \geq 2$, and let $T = \sum_{i=1}^n X_i$ be the number of successes.

1. Let $x \in \{0,1\}^n$ be a data vector. Compute the conditional probability $\mathbb{P}_\theta(X = x | X_1 = x_1)$ and show that the first observation $X_1$ on its own is not sufficient for $\theta$.

2. Using the factorisation theorem (proposition 8.3), show that $T$ is sufficient for $\theta$, confirming example 8.2.

3. Is the statistic $(X_1, T)$ sufficient for $\theta$? Justify your answer.

**Exercise 8.2.** Let $X_1, \ldots, X_n \sim \text{Uniform}(-\theta, \theta)$ be i.i.d. with $\theta > 0$, so that a single observation has density $f(x_i;\theta) = (2\theta)^{-1}\mathbf{1}_{\{-\theta \leq x_i \leq \theta\}}$.

1. Show that the joint density can be written as

$$f(x;\theta) = (2\theta)^{-n}\, \mathbf{1}_{\{\max_i |x_i| \leq \theta\}}.$$

2. Deduce from proposition 8.3 that $M = \max_i |X_i|$ is sufficient for $\theta$. Identify the functions $g$ and $h$.

3. Show that the pair $\big(X_{(1)}, X_{(n)}\big)$ of the sample minimum and maximum is also sufficient for $\theta$, and express $M$ as a function of this pair.

**Exercise 8.3.** Let $X_1, \ldots, X_n$ be i.i.d. exponential with rate $\theta > 0$, *i.e.* $f(x;\theta) = \theta e^{-\theta x}$ for $x \geq 0$.

1. Show that $T = \sum_i X_i$ is sufficient for $\theta$.
2. Using proposition 8.11, show that $T$ is minimal sufficient.
3. In example 5.2 we have found the MLE $\hat{\theta} = 1/\bar{X}$. Write $\hat{\theta}$ as a function of $T$ and explain why the factorisation in the first part of the question guarantees that such a function exists.

**Exercise 8.4.** Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ be i.i.d.

1. Assume that $\mu$ is known and $\sigma^2$ is unknown. Show that $\sum_i (X_i - \mu)^2$ is sufficient for $\sigma^2$.
2. Assume that both parameters are unknown. By expanding the squares in the exponent of the joint density, show that the joint density depends on the data only via the pair $T = (\sum_i X_i, \sum_i X_i^2)$. Using proposition 8.3, show that $T$ is sufficient for $(\mu, \sigma^2)$.
3. Using proposition 8.11, show that the pair $T$ from the previous part is minimal sufficient for $(\mu, \sigma^2)$.
4. Is the sample mean $\bar{X}$ on its own sufficient for $(\mu, \sigma^2)$ when both parameters are unknown? Justify your answer using the previous part.

**Exercise 8.5.** Let $X_1, \ldots, X_n$ be i.i.d. exponential with rate $\theta > 0$, and let $X_{(1)} = \min_i X_i$.

1. Show that $X_{(1)}$ is exponentially distributed with rate $n\theta$, first by proposition 8.8 and then directly from $\mathbb{P}_\theta(X_{(1)} > y)$.
2. Show that $nX_{(1)}$ is an unbiased estimator for the mean $1/\theta$, and compute its variance.
3. Compare the variance of $nX_{(1)}$ with that of the sample mean $\bar{X}$, which is also unbiased for $1/\theta$, and comment on the behaviour of the two estimators as $n$ grows. Is $nX_{(1)}$ a function of the sufficient statistic $\sum_i X_i$?

**Exercise 8.6.** Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability $\theta \in (0, 1)$, where $n \geq 2$, and let $T = (X_1, \ldots, X_n)$ be the whole sample.

1. Show that $w(T) = X_1 - X_2$ satisfies $\mathbb{E}_\theta(w(T)) = 0$ for all $\theta$ and $\mathbb{P}_\theta(w(T) = 0) < 1$.
2. Using definition 8.13, show that $T$ is not complete, despite $T$ being sufficient.

# Lecture 9

# Consistency by Simulation

In this lecture we use simulation to watch an estimator get closer to the truth as the sample size grows, *i.e.* the consistency from lecture 2. We see this in three ways: through a single estimate as observations are added, through the sampling distribution at three sample sizes and through the mean squared error as a function of $n$. On the way we extend the Monte Carlo skills from lecture 6 by a loop which varies the sample size. We conclude with a simulation which shows what lecture 8 means by a sufficient statistic. As before, appendix B contains a written reference for this topic.

## 9.1 A Running Estimate

In this section we simulate one long sample and watch the estimate change as the observations arrive one by one.

We will consider the exponential distribution with rate $\theta = 2$, where the maximum likelihood estimator for the rate is $\hat{\theta}_n = 1/\bar{X}_n$, as found in example 5.2. This estimator is consistent for $\theta$ by example 2.12 and proposition A.17. The most direct way to see the estimator at work is to simulate one long sample and to re-compute the estimate after each observation. The command `cumsum` can be used to compute the running totals $X_1$, $X_1 + X_2$, $X_1 + X_2 + X_3, \ldots$ in one step, and dividing the count $n$ by these totals gives all 2000 running estimates at once. Before you look at the plot, sketch the path you expect. Where is it most volatile?

```
theta   <- 2
set.seed(2026)
x       <- rexp(2000, rate = theta)
n       <- seq_len(2000)
running <- n / cumsum(x)
plot(n, running, type = "l", log = "x", ylim = c(1, 3.5),
     xlab = "n", ylab = "running estimate")
abline(h = theta, lty = 2)
abline(h = theta + c(-0.1, 0.1), lty = 3)
```

Figure 9.1 shows the expected behaviour: the path fluctuates strongly for small $n$, crosses the edge of the band $\theta \pm 0.1$ several times and ends inside it. A single path, however, cannot tell us how likely it is to be outside the band at a given $n$, and this is what consistency is about.

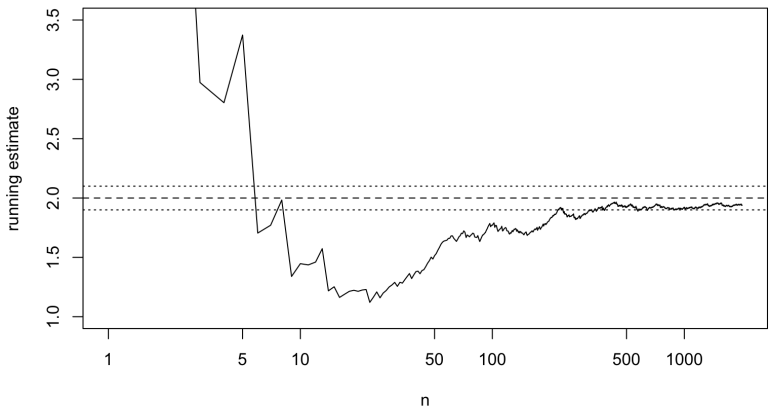



Figure 9.1: The running estimate $1/\bar{X}_n$ for the exponential rate $\theta = 2$, plotted against $n$ on a logarithmic axis. The truth is given by the dashed line, the band $\theta \pm 0.1$ is given by the dotted lines.

## 9.2 The Sampling Distribution as $n$ Grows

In this section we compare the sampling distribution at three sample sizes and which fraction of estimates misses the truth by more than a fixed amount.

By definition 2.9, consistency means that for every band around the truth, here $\theta \pm 0.1$, the probability $\mathbb{P}_\theta(|\hat{\theta}_n - \theta| \geq 0.1)$ of missing the band goes to zero as $n$ increases. A probability is a property of many repetitions, and thus, as in lecture 6, we use `replicate` to compute 10000 independent estimates, at each of the sample sizes $n = 10$, $n = 100$ and $n = 1000$. The fraction of estimates outside the band estimates the probability; as in appendix B, the mean of a condition is the fraction of cases where it holds. Before you run the code, write down three guesses for the three fractions. Will the fraction for $n = 1000$ be close to zero?

```
set.seed(2026)
est10   <- replicate(10000, 1 / mean(rexp(10, rate = theta)))
est100  <- replicate(10000, 1 / mean(rexp(100, rate = theta)))
est1000 <- replicate(10000, 1 / mean(rexp(1000, rate = theta)))
c(mean(abs(est10 - theta) >= 0.1),
  mean(abs(est100 - theta) >= 0.1),
  mean(abs(est1000 - theta) >= 0.1))
```

```
## [1] 0.8756 0.6211 0.1160
```

The probability of missing the band decreases from 0.88 at $n = 10$ to 0.12 at $n = 1000$, as consistency requires, but at $n = 1000$ it is still far from zero: the band is narrow, and consistency only says that the probability goes to zero eventually. To see the whole sampling distribution we draw the three histograms with a common horizontal axis, as in lecture 6. New is the argument `freq = FALSE`, which draws the histogram on the density scale, so that the total area of the bars is one and a narrower histogram must be a taller one. Values outside the

horizontal range are not shown; for $n = 10$ these are about 8% of the estimates, almost all of them in the long right tail. The dotted lines mark the band.

```
par(mfrow = c(3, 1))
hist(est10, breaks = 60, freq = FALSE, col = "grey85", main = "n = 10",
     xlab = "estimate", xlim = c(1, 3.5))
abline(v = theta + c(-0.1, 0.1), lty = 3)
hist(est100, breaks = 40, freq = FALSE, col = "grey85", main = "n =
    100",
     xlab = "estimate", xlim = c(1, 3.5))
abline(v = theta + c(-0.1, 0.1), lty = 3)
hist(est1000, breaks = 40, freq = FALSE, col = "grey85", main = "n =
    1000",
     xlab = "estimate", xlim = c(1, 3.5))
abline(v = theta + c(-0.1, 0.1), lty = 3)
```

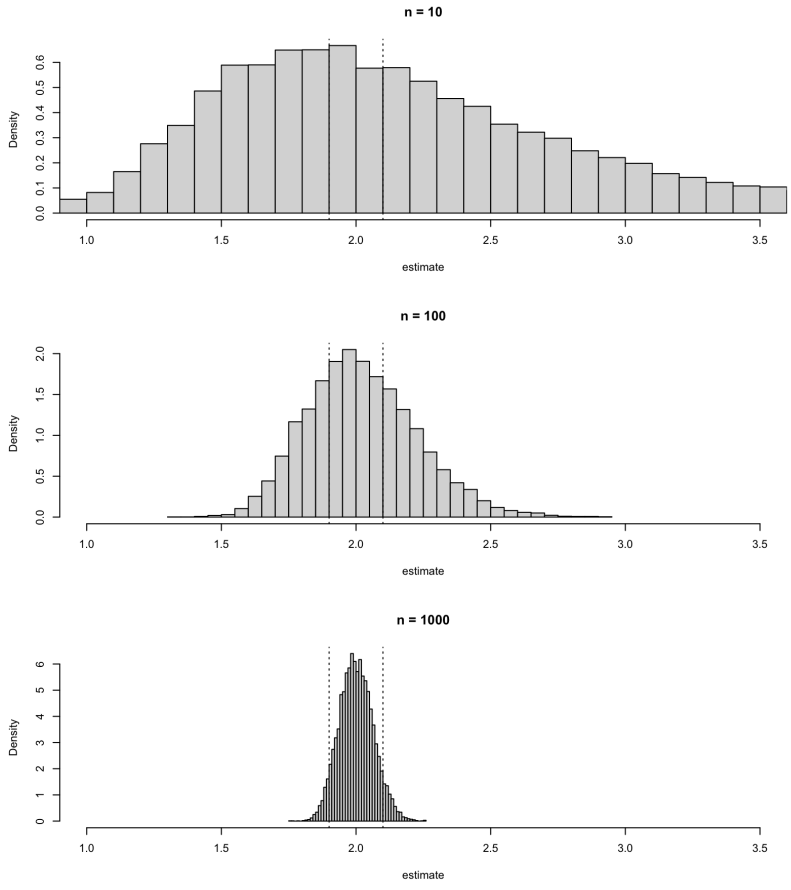



Figure 9.2: Histograms of 10000 simulated values of $1/\bar{X}_n$ for the exponential rate $\theta = 2$, at $n = 10$, $n = 100$ and $n = 1000$, on the density scale and with a common horizontal axis. The dotted lines mark the band $\theta \pm 0.1$.

Figure 9.2 is the picture promised in lectures 3 and 6: the histogram gets tighter around the truth as $n$ grows. For $n = 10$ it is also visibly skewed to the right, since $1/\bar{X}_n$ is large whenever the sample mean is small, while for $n = 1000$ it is close to symmetric.

## 9.3 The Mean Squared Error as a Function of the Sample Size

In this section we estimate the mean squared error of two estimators for a range of sample sizes, collect the results in a table and plot them.

For a report a single number per sample size is often more useful than a histogram. In lecture 6 we estimated the mean squared error of the two uniform estimators $2\bar{X}$ and $\max_i X_i$ for $n = 10$. We now repeat this study for several sample sizes, which is the usual shape of a simulation study: an outer loop over a setting, an inner Monte Carlo run and a table of results. We first put both estimates into one function, so that both are computed from the same sample and returned as a vector of length two. The command `replicate` then collects the 10000 repetitions into a *matrix*, a rectangular table of numbers with two rows, one per estimator, and 10000 columns, one per repetition, and the command `rowMeans` can be used to average each row.

```
est_both <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  c(2 * mean(sample), max(sample))
}
```

The outer loop runs over a vector `ns` of sample sizes, using `seq_along(ns)` to count through its positions. The command `matrix(0, nrow = length(ns), ncol = 2)` can be used to set aside a matrix for the results, `mse[j, ]` refers to the $j$-th row of this matrix, and row and column names, set using `rownames` and `colnames`, can be used to make the matrix a readable table. As in lecture 6, the box below shows the complete script, with the lines of the loop kept together for clarity, but with the lines in a shuffled order.

**Exercise.** Before you read the script, you need to work out the correct order of lines to make the script work.

```
signif(mse, 3)
for (j in seq_along(ns)) {
  ests    <- replicate(10000, est_both(ns[j], theta))
  mse[j, ] <- rowMeans((ests - theta)^2)
}
mse   <- matrix(0, nrow = length(ns), ncol = 2)
set.seed(2026)
rownames(mse) <- ns
theta <- 1
colnames(mse) <- c("2 * mean", "max")
ns    <- c(5, 10, 20, 50, 100, 200, 500, 1000)
```

The rule for writing R scripts is still in force: all names must be defined before they are used, so `ns` must come before the matrix is created and `theta` before the loop, the seed must be set before the loop starts, and the display (using `signif` to round the numbers) must come last. As in lecture 6, several orders satisfy these rules.

For $n = 10$ the mean squared error of the maximum was approximately half of the one for $2\bar{X}$. Before you run the reconstructed script, make a prediction. Will this ratio stay close to two for $n = 1000$?

```
theta <- 1
ns    <- c(5, 10, 20, 50, 100, 200, 500, 1000)
mse   <- matrix(0, nrow = length(ns), ncol = 2)
set.seed(2026)
for (j in seq_along(ns)) {
  ests    <- replicate(10000, est_both(ns[j], theta))
  mse[j, ] <- rowMeans((ests - theta)^2)
}
rownames(mse) <- ns
colnames(mse) <- c("2 * mean", "max")
signif(mse, 3)
```

```
##      2 * mean      max
## 5    0.065900 4.58e-02
## 10   0.033300 1.52e-02
## 20   0.016600 4.48e-03
## 50   0.006610 7.51e-04
## 100  0.003310 1.91e-04
## 200  0.001660 5.07e-05
## 500  0.000668 7.86e-06
## 1000 0.000331 1.99e-06
```

The values in each column decrease towards zero as $n$ increases, and thus both estimators are consistent by lemma 2.11. The ratio between the two columns is not close to two anymore: for $n = 1000$ the mean squared error for the maximum is smaller by a factor of approximately 170. This is expected, since the exact mean squared error $\theta^2/(3n)$ of exercise 2.2 decreases like $1/n$, while the exact value $2\theta^2/((n+1)(n+2))$ of exercise 4.4 decreases like $1/n^2$. On log-log axes, which the argument `log = "xy"` selects, a quantity proportional to $n^{-k}$ is a straight line with slope $-k$. The command `points` can be used to add a second set of points to an existing plot, with the plotting symbol chosen by the argument `pch`, and the command `legend` can be used to add a key. Will the two sets of points be parallel?

```
plot(ns, mse[, 1], log = "xy", pch = 16, xlab = "n", ylab = "MSE",
     ylim = range(mse))
points(ns, mse[, 2], pch = 1)
lines(ns, theta^2 / (3 * ns), lty = 2)
lines(ns, 2 * theta^2 / ((ns + 1) * (ns + 2)), lty = 3)
legend("bottomleft", legend = c("2 * mean", "max"),
       pch = c(16, 1), lty = c(2, 3))
```

Both sets of points lie on straight lines: the slope is $-1$ for $2\bar{X}$ and the slope is $-2$ for the maximum. The rate $1/n$ is the usual rate of decrease for a mean squared error; it is the rate of the sample mean in example 2.8. The faster rate for the maximum is a consequence of the support of the uniform distribution moving with $\theta$, the same feature which, by the remark after example 7.4 in lecture 7, keeps the uniform distribution out of the exponential families. We will consider this example again in lecture 13.

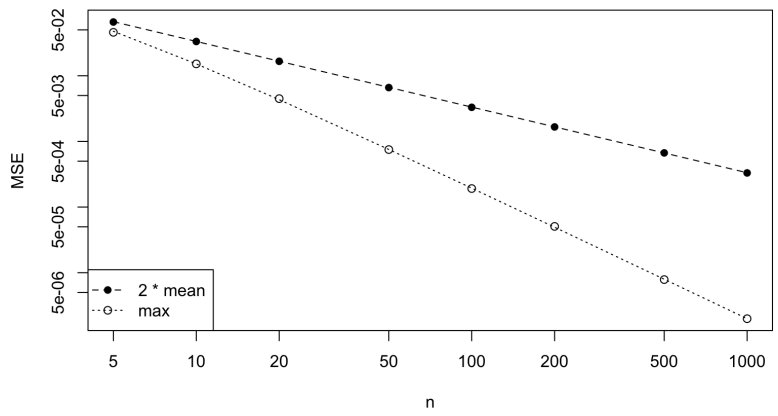



Figure 9.3: Simulated mean squared errors for $2\bar{X}$ (filled circles) and $\max_i X_i$ (open circles), for the uniform parameter $\theta = 1$, on logarithmic axes. The exact values $\theta^2/(3n)$ (dashed line) and $2\theta^2/((n+1)(n+2))$ (dotted line) are also shown.

## 9.4 A Sufficient Statistic in R

In this section we check by simulation that the conditional distribution of a Poisson sample given its total does not depend on $\theta$.

To conclude this lecture, we consider an example from lecture 8: for Poisson samples, the total $\sum_i X_i$ is sufficient by example 8.6. Be definition of sufficiency, given the total, the conditional distribution of the sample does not depend on $\theta$. We can see this by simulation, without any algebra. We generate many Poisson samples of size $n = 5$, keep only those whose total equals 10, and look at the first observation in the kept samples; we do this for two different values of $\theta$.

The command `replicate` returns a matrix with 5 rows and one column per sample, `colSums` computes the column totals, and `samples[1, totals == 10]` selects from the first row the columns where the condition holds. The command `table` can be used to count how often each value occurs, and dividing by the number of kept samples turns the counts into proportions. Before you run the code, make a prediction. Will the two tables of proportions be different, since the two values of $\theta$ are different? What will the mean of the first observation be in each case?

```
first_given_total <- function(theta, N) {
  samples <- replicate(N, rpois(5, lambda = theta))
  totals  <- colSums(samples)
  samples[1, totals == 10]
}
set.seed(2026)
first1 <- first_given_total(1.5, 100000)
first3 <- first_given_total(3, 100000)
c(length(first1), length(first3))
round(table(first1) / length(first1), 3)
round(table(first3) / length(first3), 3)
c(mean(first1), mean(first3))
```

```
## [1] 8749 4870
## first1
##     0     1     2     3     4     5     6     7
## 0.110 0.269 0.296 0.203 0.087 0.028 0.006 0.001
## first3
##     0     1     2     3     4     5     6     7
## 0.105 0.266 0.300 0.210 0.087 0.025 0.007 0.001
## [1] 2.002857 2.014168
```

The total equals 10 for about 9% of the samples when $\theta = 1.5$ and for about 5% when $\theta = 3$, so the two values of $\theta$ clearly matter for how often the total is 10. Once the total is fixed, however, the two tables agree up to simulation error, and both means are close to $10/5 = 2$: given the total, the first observation has the same distribution whether $\theta = 1.5$ or $\theta = 3$. This is definition 8.1 in action. All that the data can tell us about $\theta$ is carried by the total, and the distribution of the individual counts around it is the same for every $\theta$. (The proportions in the table are themselves estimates; with about 9000 and 5000 kept samples the simulation error is between 0.005 and 0.007 for the larger entries.)

## 9.5 A Task to Take Home

The following task is a review of the topics of this lecture, in the style of a practical report. Following the structure of section 9.3, for the exponential rate with $\theta = 2$ and sample sizes given in `ns`, estimate the bias and mean squared error of the maximum likelihood estimator $1/\bar{X}$ using 10000 repetitions each, in a table with named rows and columns. Create a plot of the mean squared error against $n$, using logarithmic axes. From the plot, determine the slope. Compare the empirical bias to the exact value from exercise 2.3. Write three sentences about what the table and plot show about the consistency of the estimator $1/\bar{X}$.

Before the next session, type in the commands from this session and check that the output matches what is shown in the text.

*Summary.*

- Appendix B is the written reference for `cumsum`, the matrices returned by `replicate`, `colSums`, `rpois`, `table` and `legend`.
- The command `cumsum` can be used to compute all running estimates of a growing sample at once. One path shows an estimate settling down, but consistency is a statement about the probability of missing a band at a fixed $n$, which only many repetitions can show.
- When the Monte Carlo study is repeated at several sample sizes, the fraction of estimates outside the band decreases and the histograms get tighter around the truth. With `freq = FALSE` a histogram is drawn on the density scale.
- A loop over a vector of sample sizes, with the results of each Monte Carlo run stored in one row of a matrix, gives the mean squared error as a function of $n$. On log-log axes a mean squared error proportional to $n^{-k}$ is a straight line with slope $-k$: for the uniform parameter, $2\bar{X}$ shows slope $-1$, the usual rate, and $\max_i X_i$ slope $-2$.
- Keeping only the Poisson samples with a given total, the distribution of the first observation is the same for two different values of $\theta$: the total is sufficient, in the sense of definition 8.1.

# Lecture 10

# Improving Estimators: Rao–Blackwell and Lehmann–Scheffe

In lecture 8 we have seen that a sufficient statistic $T$ can be used to summarise all the information in the data about the parameter. In this lecture we will use this idea to construct improved versions of estimators. The Rao–Blackwell theorem states that conditioning an unbiased estimator on the value of $T$ never increases its variance. The Lehmann–Scheffe theorem states that, if $T$ is a complete sufficient statistic, this procedure gives the unique best unbiased estimator. Finally, Basu's theorem gives the independence of $\bar{X}$ and $S^2$ for normal samples and this result will be used in lectures 25 and 29. The proofs of these three results all use facts about conditional expectations, as shown in propositions A.9 and A.11 in appendix A.

## 10.1 The Rao–Blackwell Theorem

Assume that we want to estimate a function $g(\theta)$ of the parameter, *e.g.* we could be interested in the probability $\mathbb{P}_\theta(X_1 = 0)$ of zero counts. As in lecture 2, an estimator $U$ is *unbiased* for $g(\theta)$, if $\mathbb{E}_\theta(U) = g(\theta)$ for all $\theta \in \Theta$. While unbiased estimators are often easy to find, they often turn out to be wasteful, since they only use a small amount of the available sample. The Rao–Blackwell theorem uses a sufficient statistic $T$ to improve an existing estimator by replacing $U$ with the conditional expectation $V = \mathbb{E}_\theta(U|T)$, *i.e.* the average of $U$ for all samples which have the same value of $T$ as the observed sample.

**Proposition 10.1** (Rao–Blackwell)**.** *Let $T$ be a sufficient statistic for $\theta$ and let $U$ be an unbiased estimator for $g(\theta)$ with $\mathrm{Var}_\theta(U) < \infty$ for all $\theta$. Then $V = \mathbb{E}_\theta(U|T)$ satisfies the following properties.*

1. *The random variable $V$ does not depend on $\theta$, but is a function of $T$ alone and thus a statistic.*
2. *The statistic $V$ is an unbiased estimator for $g(\theta)$.*
3. *For all $\theta \in \Theta$ we have $\mathrm{Var}_\theta(V) \leq \mathrm{Var}_\theta(U)$, with equality if and only if $\mathbb{P}_\theta(U = V) = 1$.*

*Proof.* From definition 8.1 we know that the conditional distribution of the sample $(X_1, \dots, X_n)$ given $T$ does not depend on $\theta$. Since $U$ is a function of the sample, the conditional distribution and thus the conditional expectation given $T$ do not depend on $\theta$. Thus, $V$ is a function of $T$ and by definition 1.3 a statistic.

For the second statement, using the tower property from proposition A.9, we find

$$\mathbb{E}_\theta(V) = \mathbb{E}_\theta\big(\mathbb{E}_\theta(U|T)\big) = \mathbb{E}_\theta(U) = g(\theta)$$

for all $\theta$. This shows that $V$ is unbiased.

Finally, for the third statement, we can use the law of total variance, proposition A.11, with $U$ instead of $X$ and $T$ instead of $Y$. Since $\mathbb{E}_\theta(U|T) = V$ we get

$$\mathrm{Var}_\theta(U) = \mathbb{E}_\theta\big(\mathrm{Var}_\theta(U|T)\big) + \mathrm{Var}_\theta(V).$$

Since the conditional variance $\mathrm{Var}_\theta(U|T)$ is non-negative, we find $\mathrm{Var}_\theta(V) \leq \mathrm{Var}_\theta(U)$. Equality holds, if and only if $\mathbb{E}_\theta\big(\mathrm{Var}_\theta(U|T)\big) = 0$ and since a non-negative random variable with expectation zero is zero with probability one, this is the case if and only if $\mathrm{Var}_\theta(U|T) = 0$ with probability one, *i.e.* if and only if $U$ coincides with its conditional expectation $V$ with probability one. This completes the proof. □

Since both estimators are unbiased, the mean squared errors equal the variances, as given in lemma 2.7, and thus $V$ is at least as good as $U$, and strictly better unless $U$ was already a function of $T$. The procedure of passing from $U$ to $V$ is called *Rao–Blackwellisation*. The necessity of sufficiency is highlighted by the fact that conditioning on a statistic which is not sufficient in general will result in a quantity which depends on $\theta$ but is not an estimator.

**Example 10.2.** Let $X_1, \dots, X_n$ be i.i.d. Poisson with parameter $\theta$ and assume that we want to estimate $g(\theta) = \mathbb{P}_\theta(X_1 = 0) = e^{-\theta}$. The indicator $U = \mathbf{1}_{\{X_1=0\}}$ is unbiased for $g(\theta)$, since $\mathbb{E}_\theta(U) = \mathbb{P}_\theta(X_1 = 0)$. By corollary 8.5 the sum $T = \sum_i X_i$ is sufficient. Given $T = t$, the first observation is distributed as $\mathrm{Binomial}(t, 1/n)$ and thus we have

$$V = \mathbb{E}_\theta(U|T) = \mathbb{P}_\theta(X_1 = 0|T) = \Big(1 - \frac{1}{n}\Big)^T.$$

The Rao–Blackwellised estimator now uses all $n$ observations. Exercises 10.1 and 10.2 show how to find the conditional distribution and how to apply the same procedure to a continuous model.

## 10.2 Completeness and the Lehmann–Scheffe Theorem

The Rao–Blackwell theorem improves an existing estimator, but two different unbiased estimators could in principle be improved to two different functions of $T$. The concept of completeness, definition 8.13, solves this problem: if $T$ is complete, there can only be one unbiased estimator for $g(\theta)$ among the functions of $T$ and thus all Rao–Blackwellisations will converge to this estimator. We start our discussion by naming the target.

**Definition 10.3.** An unbiased estimator $V$ for $g(\theta)$ with $\operatorname{Var}_\theta(V) < \infty$ for all $\theta \in \Theta$ is a *uniformly minimum variance unbiased estimator* (UMVUE) for $g(\theta)$, if $\operatorname{Var}_\theta(V) \leq \operatorname{Var}_\theta(W)$ for all $\theta \in \Theta$ and for every unbiased estimator $W$ for $g(\theta)$ with finite variance.

The word "uniformly" refers to the parameter: the inequality has to hold for every $\theta$ at once, and there is no reason a priori why such an estimator should exist. The following theorem shows that, whenever a complete sufficient statistic exists, a UMVUE exists as well and is easy to identify.

**Theorem 10.4** (Lehmann–Scheffe)**.** *Let $T$ be a complete sufficient statistic for $\theta$ and let $V = \varphi(T)$ be a function of $T$, an unbiased estimator for $g(\theta)$ with $\operatorname{Var}_\theta(V) < \infty$ for all $\theta$. Then $V$ is a UMVUE for $g(\theta)$. Furthermore, $V$ is the unique UMVUE: any other unbiased estimator $W$ for $g(\theta)$ with $\operatorname{Var}_\theta(W) = \operatorname{Var}_\theta(V)$ for all $\theta$ satisfies $\mathbb{P}_\theta(W = V) = 1$ for all $\theta$.*

*Proof.* To show the uniqueness of $V$ as an unbiased estimator for $g(\theta)$ which is a function of $T$, let $\psi(T)$ be another such function. Then we have $\mathbb{E}_\theta\big(\psi(T)\big) = g(\theta) = \mathbb{E}_\theta\big(\varphi(T)\big)$ for all $\theta$. The function $h = \psi - \varphi$ satisfies $\mathbb{E}_\theta\big(h(T)\big) = 0$ for all $\theta$ and, since $T$ is complete, by definition 8.13, we have $\mathbb{P}_\theta\big(h(T) = 0\big) = 1$ for all $\theta$, *i.e.* $\psi(T) = \varphi(T)$ with probability one.

Now let $W$ be any unbiased estimator for $g(\theta)$ with finite variance. Then, by the Rao–Blackwell theorem, proposition 10.1, the random variable $\mathbb{E}_\theta(W|T)$ is a function of $T$ which is unbiased for $g(\theta)$. Thus, by the first step of the proof, it equals $V$ with probability one. By the Rao–Blackwell theorem we have then

$$\operatorname{Var}_\theta(V) = \operatorname{Var}_\theta\big(\mathbb{E}_\theta(W|T)\big) \leq \operatorname{Var}_\theta(W)$$

for all $\theta$, *i.e.* $V$ is a UMVUE. If $\operatorname{Var}_\theta(W) = \operatorname{Var}_\theta(V)$ for all $\theta$, then the equality case of the Rao–Blackwell theorem implies $W = \mathbb{E}_\theta(W|T) = V$ with probability one. This completes the proof. □

The theorem turns the search for a best unbiased estimator into a two-step recipe: find a complete sufficient statistic $T$, then find any function of $T$ which is unbiased for $g(\theta)$, by adjusting a constant or by Rao–Blackwellising a crude unbiased estimator. For the models of this module the following two facts settle the completeness.

**Proposition 10.5.**

1. *Let $X_1, \ldots, X_n$ be an i.i.d. sample from an exponential family of full rank, as given in definitions 7.1 and 7.8. Then the corresponding natural statistic $T_n = \sum_{i=1}^n T(X_i)$ is complete.*

2. *Let $X_1, \ldots, X_n$ be i.i.d. and uniformly distributed on the set $(0, \theta)$ where $\theta > 0$. Then the sample maximum $\max_i X_i$ is complete.*

This proposition is given without proof. The first statement rests on the uniqueness theorem for Laplace transforms. The second statement rests on taking derivatives of the relation $\int_0^\theta h(t)\, t^{n-1}\, \mathrm{d}t = 0$ with respect to $\theta$. Both statements are allowed to be quoted. Together with corollary 8.5 and example 8.4 they form a complete sufficient statistic for all families of the form given in tables 1.1 and 1.2 (with $m$ known for the binomial). If we use a one-to-one function of a complete sufficient statistic, the new function is again complete and sufficient. Thus we can use the function $\bar{X}$ instead of $\sum_i X_i$ if required.

**Example 10.6.** For the Poisson sample from example 10.2, the natural parameter $\eta = \log\theta$ takes values on all of $\mathbb{R}$ and thus the sum $T = \sum_i X_i$ is a complete and sufficient statistic. Since $\bar{X} = T/n$ is a function of $T$ and is unbiased for $\theta$, it is the UMVUE for $\theta$ by theorem 10.4. Similarly, the Rao–Blackwellised indicator $V = (1 - 1/n)^T$ is the UMVUE for $e^{-\theta}$.

**Example 10.7.** Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ with both parameters unknown. This is a two-parameter exponential family with natural statistic $T(x) = (x, x^2)$, and the pair $(\bar{X}, S^2)$ is sufficient by example 8.7. The family has full rank, and thus the pair is also complete; exercise 10.4 verifies the rank condition. Since $\bar{X}$ and $S^2$ are unbiased for $\mu$ and $\sigma^2$ by lemmas 2.2 and 2.3, they are the UMVUEs for these parameters.

The second example puts section 2.3 into perspective: there the biased estimator $\frac{1}{n+1}\sum_i (X_i - \bar{X})^2$ had smaller mean squared error than $S^2$. No contradiction, since the Lehmann–Scheffe theorem compares $S^2$ to unbiased estimators only.

## 10.3 Ancillary Statistics and Basu's Theorem

A sufficient statistic contains all the information about $\theta$. At the opposite extreme are statistics which contain none.

**Definition 10.8.** A statistic $A = A(X_1, \dots, X_n)$ is *ancillary* for $\theta$, if the distribution of $A$ under $\mathbb{P}_\theta$ does not depend on $\theta \in \Theta$.

The running example is a normal sample $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with known variance $\sigma^2$ and unknown mean $\mu$. Writing $Z_i = X_i - \mu$ we have $X_i - \bar{X} = Z_i - \bar{Z}$ and the $Z_i$ are i.i.d. $N(0, \sigma^2)$, for every value of $\mu$. Thus, the vector of residuals $(X_1 - \bar{X}, \dots, X_n - \bar{X})$ is ancillary for $\mu$ and every function of the residuals is ancillary, *e.g.* the sample variance $S^2$ and the range $\max_i X_i - \min_i X_i$; we will consider the range in exercise 10.5.

One might expect an ancillary statistic to have nothing to do with a sufficient one. Under completeness this becomes a theorem, and it is a surprisingly effective way of proving independence.

**Proposition 10.9** (Basu)**.** *Let $T$ be a complete sufficient statistic for $\theta$ and let $A$ be an ancillary statistic. Then $T$ and $A$ are independent of each other under $\mathbb{P}_\theta$ for all $\theta \in \Theta$.*

*Proof.* Let $B$ be a set of possible values of $A$. Since $A$ is ancillary, the probability $p_B = \mathbb{P}_\theta(A \in B)$ does not depend on $\theta$. We compare this probability to the conditional probability

$$h(T) = \mathbb{P}_\theta(A \in B \,|\, T) = \mathbb{E}_\theta\big(\mathbf{1}_{\{A \in B\}} \,\big|\, T\big).$$

Since $A$ is a function of the sample and since $T$ is sufficient, the conditional distribution of $A$ given $T$ does not depend on $\theta$. Thus, $h(T)$ is a function of $T$ alone. Using the tower property from proposition A.9 we find

$$\mathbb{E}_\theta\big(h(T)\big) = \mathbb{P}_\theta(A \in B) = p_B$$

for all $\theta$. Thus the function $h - p_B$ satisfies $\mathbb{E}_\theta\big(h(T) - p_B\big) = 0$ for all $\theta$. Since $T$ is complete, we have $h(T) = p_B$ with probability one for every $\mathbb{P}_\theta$.

It remains to show that this implies independence. For any set $C$ of possible values of $T$, the indicator function $\mathbf{1}_{\{T\in C\}}$ is a function of $T$ and can be taken out of the conditional expectation given $T$ (see proposition A.10 in appendix A). Using the tower property we find

$$\begin{aligned}\mathbb{P}_\theta(T \in C,\ A \in B) &= \mathbb{E}_\theta(\mathbf{1}_{\{T\in C\}}\mathbf{1}_{\{A\in B\}}) = \mathbb{E}_\theta\Big(\mathbf{1}_{\{T\in C\}}\,\mathbb{E}_\theta(\mathbf{1}_{\{A\in B\}}\,|\,T)\Big)\\ &= \mathbb{E}_\theta(\mathbf{1}_{\{T\in C\}}\,p_B) = \mathbb{P}_\theta(T\in C)\,\mathbb{P}_\theta(A\in B).\end{aligned}$$

Since $B$ and $C$ were arbitrary, we have shown the independence of $T$ and $A$ and the proof is complete. □

The proof uses each hypothesis exactly once: ancillarity makes the unconditional probability constant in $\theta$, sufficiency makes the conditional probability a statistic, and completeness forces these two to be the same. The most important application of this result in this module is the following.

**Corollary 10.10.** *Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$. Then $\bar{X}$ and $S^2$ are independent.*

*Proof.* Fixing the value of $\sigma^2$, we can consider $\mu \in \mathbb{R}$ to be the only parameter. The density of a single observation is

$$f(x;\mu) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-x^2/(2\sigma^2)} \exp\Big(\frac{\mu}{\sigma^2}\,x - \frac{\mu^2}{2\sigma^2}\Big).$$

This is of the form given in definition 7.1 with natural statistic $T(x) = x$ and natural parameter $\eta(\mu) = \mu/\sigma^2$. As $\mu$ ranges over $\mathbb{R}$, so does $\eta(\mu) = \mu/\sigma^2$, that is, the natural parameter takes all values in $\mathbb{R}$, and thus the family is of full rank. By corollary 8.5 and proposition 10.5, the sum $\sum_i X_i$, and thus $\bar{X}$, is a complete and sufficient statistic for $\mu$. Furthermore, $S^2$ is a function of the residuals $X_i - \bar{X}$ and thus, as shown above, is ancillary for $\mu$. By Basu's theorem, proposition 10.9, the statistics $\bar{X}$ and $S^2$ are independent under $\mathbb{P}_{\mu,\sigma^2}$ for every $\mu$ and, since $\sigma^2$ was arbitrary, the claim is proved. □

This completes the proof of the independence statement of proposition A.4. The result from lectures 25 and 29 on the ratio of $\bar{X}-\mu$ to $S$ uses this proposition. The proof contains a useful trick: when both parameters are unknown, neither $\bar{X}$ nor $S^2$ can be ancillary and we cannot apply Basu's theorem directly. The solution to this problem is to fix the value of $\sigma^2$, *i.e.* to consider the independence statement for one distribution at a time.

In lecture 13 we will see a second way to derive optimality, using the Cramer–Rao bound, which does not require conditioning.

*Summary.*

- Rao–Blackwell: If an unbiased estimator is conditioned on a sufficient statistic $T$, the conditioned estimator is unbiased and has no larger variance.
- A UMVUE is an unbiased estimator which has minimal variance among all unbiased estimators, for all $\theta$.
- Lehmann–Scheffe: If $T$ is complete and sufficient, then the only function of $T$ which is unbiased for $g(\theta)$ is the UMVUE for $g(\theta)$.
- The natural statistic for a full-rank exponential family and the maximum of a uniform sample are complete, but without proof.
- Basu: A complete sufficient statistic is independent of all ancillary statistics. Using this result, one can show that $\bar{X}$ and $S^2$ of a normal sample are independent.

**Exercise 10.1.** Let $X_1, \dots, X_n$ be i.i.d. Poisson with parameter $\theta > 0$, let $T = \sum_{i=1}^n X_i$ and let $U = \mathbf{1}_{\{X_1=0\}}$, as in example 10.2. You may use the fact that a sum of independent Poisson random variables is again Poisson, with the parameters added.

1. Show that $U$ is an unbiased estimator for $e^{-\theta}$ and determine the variance of this estimator.

2. Show that for $t \in \mathbb{N}_0$ and $x \in \{0, 1, \dots, t\}$ we have

$$\mathbb{P}_\theta(X_1 = x | T = t) = \binom{t}{x} \Big(\frac{1}{n}\Big)^x \Big(1 - \frac{1}{n}\Big)^{t-x},$$

   *i.e.* that the conditional distribution of $X_1$ given $T = t$ is Binomial$(t, 1/n)$ and that this distribution does not depend on $\theta$.

3. Deduce that $V = \mathbb{E}_\theta(U|T) = (1 - 1/n)^T$.

4. Show directly that $V$ is an unbiased estimator for $e^{-\theta}$, using the probability generating function (7.3) of the Poisson distribution with parameter $n\theta$, which is given by $\mathbb{E}_\theta(z^T) = \exp(n\theta(z-1))$ (exercise 7.6).

5. Explain why $V$ is the UMVUE for $e^{-\theta}$ and discuss the differences between this UMVUE and the MLE $e^{-\bar{X}}$, introduced in exercise 5.7.

**Exercise 10.2.** Let $n \geq 2$ and let $X_1, \dots, X_n$ be i.i.d. exponential with rate $\theta > 0$, *i.e.* $f(x;\theta) = \theta e^{-\theta x}$ for $x \geq 0$ and let $T = \sum_{i=1}^n X_i$. You may use the fact that the sum of $k$ independent random variables of this type has density $f_k(t) = \theta^k t^{k-1} e^{-\theta t}/(k-1)!$ for $t \geq 0$.

1. Explain why $X_1$ is an unbiased estimator for the mean $1/\theta$ and why $T$ is sufficient for $\theta$.

2. Let $R = \sum_{i=2}^{n} X_i$. Use the independence of $X_1$ and $R$ to find the joint density of $(X_1, T)$. Deduce the conditional density of $X_1$ given $T = s$. The result is
$$f(x|s) = \frac{(n-1)(s-x)^{n-2}}{s^{n-1}} \qquad \text{for } 0 < x < s,$$
and zero elsewhere. Check that this density does not depend on $\theta$.

3. Show that $\mathbb{E}_\theta(X_1|T = s) = s/n$ and conclude that Rao–Blackwellising $X_1$ leads to the sample mean $\bar{X}$, again unbiased for $1/\theta$.

4. Find a second argument for $\mathbb{E}_\theta(X_1|T) = T/n$, based on the symmetry of the sample, without any integration.

**Exercise 10.3.** Let $X_1, \ldots, X_n$ be i.i.d. uniformly distributed on the set $(0, \theta)$ where $\theta > 0$, and let $M = \max_i X_i$.

1. Using the mean of $M$ from exercise 2.2, show that $\hat{\theta} = (n+1)M/n$ is an unbiased estimator for $\theta$.

2. Show that $\hat{\theta}$ is the UMVUE for $\theta$.

3. The estimator $\tilde{\theta} = 2\bar{X}$ is also unbiased for $\theta$. Determine the variances of $\hat{\theta}$ and of $\tilde{\theta}$ and verify that they are ordered as in theorem 10.4.

4. Without computing any conditional expectations, find $\mathbb{E}_\theta(\tilde{\theta}|M)$.

**Exercise 10.4.** Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$, where both parameters are unknown. This is the situation of example 10.7.

1. Write the density of an observation in the form from definition 7.6 and identify the natural statistic $T(x) = (x, x^2)$ and the natural parameter $\eta(\mu, \sigma^2)$.

2. Show that the family of densities has full rank, in the sense of definition 7.8, by determining the range of $\eta$.

3. Deduce that $\left(\sum_i X_i, \sum_i X_i^2\right)$ is complete and sufficient and argue that the same holds for the pair $(\bar{X}, S^2)$.

**Exercise 10.5.** Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ where $\sigma^2$ is known and $\mu \in \mathbb{R}$ is unknown. Furthermore, let $R = \max_i X_i - \min_i X_i$ be the sample range.

1. Show that $R$ is ancillary for $\mu$.

2. Show that $\bar{X}$ and $R$ are independent.

3. Now assume that $\mu$ is known and $\sigma^2 > 0$ is unknown. Is $R$ ancillary for $\sigma^2$? Is $R/\sigma$ a statistic?

# Bibliography


Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26):e2422633122, 2025. doi: 10.1073/pnas.2422633122.

W. J. Braun and D. J. Murdoch. *A First Course in Statistical Programming with R*. Cambridge University Press, Cambridge, 3rd edition, 2021.

G. Casella and R. L. Berger. *Statistical Inference*. Duxbury, Pacific Grove, 2nd edition, 2002.

D. R. Cox and D. V. Hinkley. *Theoretical Statistics*. Chapman and Hall, London, 1974.

A. C. Davison. *Statistical Models*. Cambridge University Press, Cambridge, 2003.

M. H. DeGroot and M. J. Schervish. *Probability and Statistics*. Addison-Wesley, Boston, 4th edition, 2012.

A. Gelman, J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari, and D. B. Rubin. *Bayesian Data Analysis*. Chapman and Hall/CRC, Boca Raton, 3rd edition, 2013.

G. H. Givens and J. A. Hoeting. *Computational Statistics*. Wiley, Hoboken, 2nd edition, 2013.

Mark Hodds, Lara Alcock, and Matthew Inglis. Self-explanation training improves proof comprehension. *Journal for Research in Mathematics Education*, 45(1):62–101, 2014. doi: 10.5951/jresematheduc.45.1.0062.

P. M. Lee. *Bayesian Statistics: An Introduction*. Wiley, Chichester, 4th edition, 2012.

E. L. Lehmann and G. Casella. *Theory of Point Estimation*. Springer, New York, 2nd edition, 1998.

J. A. Rice. *Mathematical Statistics and Data Analysis*. Duxbury, Belmont, 3rd edition, 2007.

David Strömberg, Victor Lei, and Yanhui Wu. The generative AI learning penalty: Evidence from Chinese secondary education. Discussion Paper DP21577, Centre for Economic Policy Research, London, 2026.

G. A. Young and R. L. Smith. *Essentials of Statistical Inference*. Cambridge University Press, Cambridge, 2005.

# Index


ancillary statistic, 67

Basu's theorem, 67

complete statistic, 53
    examples, 66
completeness, 53
cumulant, 43
cumulant function, 42
    moments of the natural statistic, 43
cumulant generating function, 42
curved exponential family, 41

estimator
    maximum likelihood, 25
    uniformly minimum variance unbiased, 66
exponential family, 39
    completeness, 66
    curved, 41
    full rank, 42
    $k$-parameter, 41
    mean and variance, 43
    one-parameter, 39
    sample from, 40
    shape, 39–40
    sufficient statistic, 51

factorisation theorem, 49
Fisher–Neyman factorisation theorem, 49
full rank, 42

index, 308
indicator function, 50
invariance of the MLE, 29

Lehmann–Scheffe theorem, 66
likelihood equation, 25
log-odds, 40
log-partition function, *see* cumulant function

maximum likelihood estimator, 25
    invariance, 29
minimal sufficient statistic, 52
    criterion, 52
MLE, *see* maximum likelihood estimator
moment generating function, 42

natural parameter, 39
natural parameter space, 42
natural parametrisation, 42
natural statistic, 39
normal distribution
    independence of sample mean and variance, 68

order statistics
    maximum and minimum, 51

probability generating function, 43

Rao–Blackwell theorem, 64
    Rao–Blackwellisation, 65
Rao–Blackwellisation, 65

statistic
    ancillary, 67
    complete, 53
    minimal sufficient, 52
    natural, 39
    sufficient, 48
sufficient statistic, 48
    factorisation theorem, 49
    in an exponential family, 51
    minimal, 52

UMVUE, *see* uniformly minimum variance unbiased estimator
uniformly minimum variance unbiased estimator, 66
    Lehmann–Scheffe theorem, 66